

IDENTIFICATION

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PRODUCT NAME: RK8E/RK8L DISK FORMATTER PROGRAM
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1. ABSTRACT

THE RK8E/RK8L DISK FORMATTER PROGRAM IS DESIGNED TO WRITE AND CHECK THE FORMAT OF THE COMPLETE DISK CARTRIDGE.

ONLY STANDARD DEC SURFACE FORMAT IS AVAILABLE (I.E. SECTORS NUMBERED IN THE NORMAL NUMERICAL SEQUENCE 0, 1, 2, 3, 4, 5, ETC.).

2. RESTRICTIONS

THE RK8L CONTROL, WHICH CAN CONTROL UP TO 8 DRIVES, WILL NOT RUN WITH THE DW8E BUS ADAPTER. THE REASON FOR THIS STATEMENT IS THAT THE RK8L CONTROL USES IOT0 FOR EXTENDED DRIVES 4-7 WHICH IS NOT AVAILABLE ON THE DW8E.

2.1 HARDWARE

- A. PDP-8/E, 8/F, 8/M OR 8/A COMPUTER
OTHER FAMILY OF 8 COMPATIBLE COMPUTER WITH NECESSARY DW8E BUS ADAPTER FOR RK8E CONTROL ONLY.
- B. AT LEAST 4K OF READ/WRITE MEMORY. AT LEAST 8K OF MEMORY IS NEEDED FOR OPERATION OF THE CONSOLE PACKAGE.
- C. ASR-33 TELETYPE OR EQUIVALENT
- D. RK8E DISK CONTROL OR RK8L DISK CONTROL
- E. RK05J OR RK05F DISK DRIVE(S)

NOTE: THE RK05F'S DRIVE IS CONSIDERED AS TWO SEPARATE UNITS. WHEN ANSWERING ALL QUESTIONS EACH SEPARATE UNIT MUST BE SPECIFIED: DSK07, DSK17, DSK27, ETC.

2.2 PROGRAM STORAGE

THE PROGRAM UTILIZES OR OCCUPIES LOCATIONS 0000 TO 4177 OF THE CURRENT FIELD.

3. PRELIMINARY PROGRAMS

THE FOLLOWING PROGRAMS SHOULD BE RUN IF THE FORMATTER PROGRAM FAILS TO OPERATE CORRECTLY:

ALL BASIC AND EXTENDED MEMORY DIAGNOSTICS

FOR THE RK8E CONTROL, RUN THE RK8E DISKLESS CONTROL TEST AND THE RK8E DRIVE CONTROL TEST.

FOR THE RK8L CONTROL, RUN THE RK8L INSTRUCTION TEST.

4. OPERATOR AND/OR PROGRAM ACTION

4.1 STANDARD TEST PROCEDURE

- A. LOAD THE PROGRAM INTO ANY R/W MEMORY BANK USING THE STANDARD BINARY LOADER TECHNIQUE.
- B. TO RUN THE FORMATTEN PROGRAM, FOLLOW THE PROCEDURE IN SECTION 4.4.

4.2 RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE

THE FOLLOWING IS THE CORRECT CARTRIDGE MOUNTING PROCEDURE FOR THE RK05J DISK DRIVE. ANY DEVIATION ENCOUNTERED DURING THIS PROCEDURE WILL BE CONSIDERED AN ERROR CONDITION.

- A. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION.
- B. TURN AC POWER ON.
- C. VERIFY THAT LIGHT LABELED "PWR" IS ON.
- D. WAIT FOR LIGHT LABELED "LOAD" TO COME ON.
- E. VERIFY THAT LIGHTS LABELED "RDY", "ON CYL", "FAULT", "WT", AND "RD" ARE OFF.
- F. OPEN ACCESS DOOR.
- G. INSERT CARTRIDGE.
- H. CLOSE ACCESS DOOR.
- I. SET SWITCH LABELED "RUN/LOAD" TO THE "RUN" POSITION.
- J. WAIT FOR LIGHTS LABELED "RDY" AND "ON CYL" TO COME ON.
- K. TOGGLE SWITCH LABELED "WT PROT" AND VERIFY THAT THE LIGHT LABELED "WT PROT" GOES ON AND OFF.
- L. TOGGLE SWITCH LABELED "WT PROT" UNTIL LIGHT LABELED "WT PROT" GOES OFF.
- M. VERIFY THAT LIGHTS LABELED "FAULT", "WT", "RD", AND "LOAD" ARE OFF.

4.3 RK05F DRIVE SETUP PROCEDURE

THE FOLLOWING IS THE CORRECT DRIVE SETUP PROCEDURE FOR THE RK05F DISK DRIVE. ANY DEVIATION ENCOUNTERED DURING THIS PROCEDURE WILL BE CONSIDERED AN ERROR CONDITION.

- A. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION.
- B. TURN AC POWER ON.

- C. VERIFY THAT LIGHT LABELED "PWR" IS ON.
- D. WAIT FOR LIGHT LABELED "LOAD" TO COME ON.
- E. VERIFY THAT LIGHTS LABELED "RDY", "ON CYL", "FAULT", "WT", AND "ND" ARE OFF.
- F. SET SWITCH LABELED "RUN/LOAD" TO THE "RUN" POSITION.
- G. WAIT FOR LIGHTS LABELED "RDY" AND "ON CYL" TO COME ON.
- H. TOGGLE SWITCH LABELED "WT PROT" AND VERIFY THAT THE LIGHT LABELED "WT PROT" GOES ON AND OFF.
- I. TOGGLE SWITCH LABELED "WT PROT" UNTIL LIGHT LABELED "WT PROT" GOES OFF.
- J. VERIFY THAT LIGHTS LABELED "FAULT", "WT", "ND", AND "LOAD" ARE OFF.

4.4 FORMAT PROGRAM -----

- A. MAKE READY ALL DRIVES TO BE FORMATTED:

 FOR RK05J DRIVES USE THE RK05 DRIVE MOUNTING PROCEDURE
 REFER TO SECTION 4.2.

 FOR RK05F DRIVES USE THE RK05 DRIVE SETUP PROCEDURE
 REFER TO SECTION 4.3.
- B. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION ON ALL DRIVES NOT BEING FORMATTED.
- C. SET THE SWITCH REGISTER TO 0200 AND PRESS LOAD ADDRESS.
- D. SET THE SWITCH REGISTER TO 0000.
- E. IF IT IS DESIRED TO CHANGE THE IOT DEVICE CODES WITHIN THE PROGRAM (THEY ARE NORMALLY X74X), SET SWITCH REGISTER BIT 11 TO A "1".
- F. IF CHANGE IOT CODES WAS SELECTED, SET SWITCH REGISTER BITS 3 TO 8 TO THE DESIRED IOT DEVICE CODE.
- G. PRESS KEY START (KEY START IS KEY CLEAR AND THEN KEY CONTINUE ON A PDP8/E, PDP8/F, OR PDP8/M). IF SELECTING A PDP8/A (PRESS INIT AND THEN PRESS RUN). IF SELECTED, ALL IOT DEVICE CODES WITHIN THE PROGRAM WILL BE CHANGED. THE TTY WILL TYPE THE FOLLOWING PROGRAM NAME, INFORMATION, AND QUESTION.

RK05/RK06 DISK FORMATTER PROGRAM

FOR ALL QUESTIONS ANSWER Y FOR YES OR N FOR NO,
FORMAT DISK 0?

- H. IF THE OPERATOR DESIRES TO FORMAT DISK 0, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

FORMAT DISK 1?

- I. IF THE OPERATOR DESIRES TO FORMAT DISK 1, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

FORMAT DISK 2?

- J. IF THE OPERATOR DESIRES TO FORMAT DISK 2, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

FORMAT DISK 3?

- K. IF THE OPERATOR DESIRES TO FORMAT DISK 3, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

FORMAT DISK 4?

- L. IF THE OPERATOR DESIRES TO FORMAT DISK 4, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

FORMAT DISK 5?

- M. IF THE OPERATOR DESIRES TO FORMAT DISK 5, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

FORMAT DISK 6?

- N. IF THE OPERATOR DESIRES TO FORMAT DISK 6, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

FORMAT DISK 7?

- O. IF THE OPERATOR DESIRES TO FORMAT DISK 7, TYPE Y FOR YES, OTHERWISE, N FOR NO, ON THE TTY KEYBOARD. THE FOLLOWING QUESTION WILL THEN BE TYPED ON THE TTY.

ARE YOU SURE?

- P. TYPING N FOR NO WILL RESULT IN REPEATING ALL THE PREVIOUS QUESTIONS. TYPING Y FOR YES, WILL RESULT IN EXECUTION OF THE OPERATION SELECTED.

- Q. PROGRAM EXECUTION IS APROX. 80 SECONDS PER DISK DRIVE. AFTER ALL DISKS SELECTED HAVE BEEN FORMATTED AND CHECKED THE TTY WILL TYPE THE FOLLOWING PASS COMPLETE MESSAGE AND

QUESTION,

RK8E/RK8L DISK FORMATTER PASS COMPLETE
FORMAT SAME DISK(S) AGAIN?

R. IF THE OPERATOR DESIRES TO REPEAT THE OPERATION SELECTED,
TYPE Y FOR YES. TYPING N FOR NO WILL RESULT IN A REPEAT
OF THE INITIAL START-UP QUESTIONS.

4,5 SWITCH REGISTER SETTINGS

SWR11=0 DO NOT CHANGE IOT DEVICE CODES
SWR11=1 CHANGE IOT DEVICE CODES
SWR3=8 DESIRED IOT DEVICE CODE.

5, ERRORS

WHEN A RECOVERABLE ERROR OCCURS THE TTY WILL PRINT
AN "ERROR HEADER" AND ERROR INFORMATION PERTAINING
TO THE FAILURE.

POSSIBLE ERROR HEADERS ARE AS FOLLOWS.

DISK DATA ERROR
READ STATUS ERROR
WRITE STATUS ERROR
RECALIBRATE STATUS ERROR

AFTER THE ERROR HEADER MENTIONED ABOVE IS TYPED THE TTY
WILL PRINT SOME OF THE FOLLOWING ERROR INFORMATION PER-
TAINING TO THE FAILURE.

PC: PROGRAM LOCATION OF FAILURE
GD: EXPECTED INFORMATION
EX: EXTENDED DRIVE BIT
CM: SOFTWARE COMMAND REGISTER
ST: CONTENTS OF STATUS REGISTER
DA: SOFTWARE CYLINDER, SURFACE, AND SECTOR REGISTER
CA: INITIAL CURRENT ADDRESS
AD: ADDRESS OF DATA BREAK
DT: DATA FOUND DURING DATA BREAK

AFTER THE ERROR INFORMATION IS TYPED THE TTY WILL TYPE ONE
OF THE FOLLOWING QUESTIONS ASKING THE ERROR RECOVERY DESIRED.

A. IF THE ERROR WAS A RECALIBRATE ERROR THE FOLLOWING QUESTION

WILL BE TYPED.

TRY TO RECALIBRATE SAME DISK AGAIN?

TYPING A Y FOR YES WILL RESULT IN A REPEAT OF THE RE-CALIBRATE SEQUENCE ON THE DISK IN ERROR. TYPING N FOR NO WILL RESULT IN PROGRESSING TO THE NEXT AVAILABLE DISK.

B. IF THE ERROR WAS A WRITE ERROR THE FOLLOWING QUESTION WILL BE TYPED.

TRY TO FORMAT SAME CYLINDER AGAIN?

TYPING Y FOR YES WILL RESULT IN A REPEAT OF THE WRITE SEQUENCE ON THE CURRENT CYLINDER. TYPING N FOR NO WILL RESULT IN PROGRESSING TO THE NEXT SEQUENTIAL CYLINDER.

C. IF THE ERROR WAS A READ OR CHECK ERROR THE FOLLOWING QUESTION WILL BE TYPED.

TRY TO CHECK SAME CYLINDER AGAIN?

TYPING A Y FOR YES WILL RESULT IN A REPEAT IN THE READ AND CHECK SEQUENCE ON THE CURRENT CYLINDER. TYPING A N FOR NO WILL RESULT IN PROGRESSING TO THE NEXT SEQUENTIAL CYLINDER.

6. PROGRAM DESCRIPTION

THE FORMATTING IS ACTUALLY A FUNCTION OF THE RK8E OR RK8L CONTROL AND DRIVE LOGIC. THE PROGRAM SIMPLY WRITES DATA ON EVERY SECTOR IN THE "WRITE ALL" MODE, THEN CHECKS THE DATA IN SUCH A WAY IN THE "READ DATA" MODE AS TO VERIFY THAT THE HEADER WORDS WRITTEN ON EVERY SECTOR ARE ALSO CORRECT. THE "READ DATA MODE" AUTOMATICALLY PERFORMS A CHECK HEADER FUNCTION.

THE FIRST TWO WORDS OF EVERY SECTOR ARE SET TO THE ABSOLUTE DISK ADDRESS(I.E. COMMAND REGISTER BITS 9-11 AND CYLINDER, SURFACE, AND SECTOR BITS 0-11, RESPECTIVELY) AND THE REMAINDER OF THE DATA AREA TO ALL ZEROS WHEN THE DATA IS WRITTEN. ONLY THE FIRST TWO WORDS OF EVERY SECTOR(I.E. THE ADDRESSING INFORMATION) ARE CHECKED WHEN DATA IS READ IN THE "READ DATA" MODE.

7. APT-8 HOOKS

7.1 DESCRIPTION

TWO INTERFACES HAVE BEEN PROVIDED WHICH ALLOW THIS DIAGNOSTIC TO RUN UNDER THE STANDARD APT-8 SYSTEM. THESE INTERFACES ARE:

1. TIMING INTERFACE

2. ERROR INTERFACE

EACH WILL BE EXPLAINED IN DETAIL.

7.2 SETUP -----

ONLY HARDWARE CONFIGURATION WORD 2, ADDRESS 22, NEED BE ESTABLISHED. THE FOLLOWING INFORMATION MUST BE INDICATED:

1. SINGLE OR MULTIPLE DRIVE TESTING.
2. DRIVE OR DRIVES TO BE TESTED.
3. DIAGNOSTIC RUNNING UNDER APT-8.

IF SINGLE DRIVE TESTING BIT 5 OF ADDRESS 22 MUST BE SET TO A ONE (1) WITH BITS 6-11 CONTAINING THE DRIVE TO BE TESTED. IF MULTIPLE DRIVES ARE TO BE DONE BIT 5 MUST BE SET TO A ZERO (0) AND BIT 6-11 CONTAINING THE HIGHEST NUMBER DRIVE TO BE TESTED. WHEN MULTIPLE DRIVE TESTING ONLY A SPECIFIC NUMBER OF DRIVES CAN BE INDICATED. THE PROGRAM ASSUMES THE DRIVES ARE TO BE DONE BEGINNING WITH DRIVE ZERO (0) AND FINISHING WITH THE HIGHEST DRIVE INDICATED. IF MULTIPLE DRIVES OTHER THAN CONSECUTIVELY NUMBERED DRIVES BEGINNING WITH DRIVE ZERO (0) ARE TO BE DONE, THEY MUST BE DONE AS SINGLE DRIVES AND TESTED INDEPENDENTLY.

THE PROGRAM ALLOWS ONLY DRIVES ZERO (0) THROUGH SEVEN (7) TO BE TESTED AT THIS TIME.

BIT ZERO OF ADDRESS 22 MUST BE SET TO A ONE TO INDICATE THAT THE PROGRAM WILL RUN UNDER APT-8.

NOTE: IT SHOULD BE NOTED AT THIS TIME THAT WHILE RUNNING UNDER APT-8 THE HARDWARE SWITCH REGISTER IS INOPERATIVE, ONLY THE HALT AND SINGLE STEP SWITCH WILL EFFECT THE PROGRAM RUN.

7.3 APT-8 INTERFACES -----

7.3.1. TIMING -----

APT-8 IS NOTIFIED OF PROGRAM RUN BETWEEN .2 SEC AND 2.0 SEC ON A 1.2 MICROSECOND MEMORY CYCLE. THIS WILL ALLOW THE DIAGNOSTIC TO RUN WITHOUT CAUSING AN APT-8 TIMEOUT ERROR IF THE DIAGNOSTIC IS TO BE RUN ON THE SLOWER MOS MEMORY.

7.3.2. ERRORS -----

ONLY THE ERROR PC IS REPORTED TO APT-8 SYSTEM. ERRORS WHICH CAUSE A PROGRAMMED HALT CAUSE A TIMEOUT ERROR. IF A PROGRAMMED HALT SHOULD OCCUR, THE ERROR PC WILL APPEAR IN THE AC ON THE DEVICE UNDER TEST. PROGRAMMED HALTS ARE EXPLAINED EARLIER IN THIS DOCUMENT.

8. PROGRAM LISTING

9. CONSOLE PACKAGE ADDENDUM

9.1 DESCRIPTION

THE CONSOLE PACKAGE HAS BEEN ADDED TO THIS DIAGNOSTIC TO ALLOW THE PROGRAM TO RUN WITH NO HARDWARE SWITCH REGISTER AND TO HAVE COMMUNICATIONS WITH THE DIAGNOSTIC VIA A TERMINAL. THE DIAGNOSTIC CAN BE RUN IN TWO MODES WITH THE CONSOLE PACKAGE . 1) RUNNING WITH THE CONSOLE PACKAGE ACTIVE - THIS ALLOWS THE OPERATOR CONTROL OF THE DIAGNOSTIC THROUGH THE TERMINAL. THE DIAGNOSTIC WILL ASK FOR THE VALUE OF THE PSEUDO SWITCH REGISTER, BEFORE CONTINUING WITH EXECUTION OF THE DIAGNOSTIC. ALL ERRORS AND PASS COMPLETES WILL BE PRINTED AT THE TERMINAL. NO HALTS WILL BE EXECUTED. 2) CONSOLE PACKAGE NOT ACTIVE-THIS WILL RESULT IN THE NORMAL STANDALONE OPERATION OF THE PROGRAM AS DISCRIBED IN SECTIONS 1 THROUGH 8 OF THIS DUCUMENT.

9.2 RESTRICTIONS

- 1) RUNNING THE CONSOLE PACKAGE REQUIRES THAT THE PSEUDO SWITCH REGISTER BE USED.
- 2) ONCE RUNNING THE CONSOLE PACKAGE NONACTIVE AND NOW DESIRE TO RUN IT ACTIVE, ONE MUST RELOAD THE DIAGNOSTIC AND INITIALIZE FOR A ACTIVE CONSOLE PACKAGE.

9.3 INITIALIZATION

FOR A ACTIVE CONSOLE PACKAGE

- 1.) SET LOCATION 21 BIT0=0 TO INDICATE USE OF PSEUDO SWITCH REGISTER.
- 2.) SET LOCATION 22 BIT3=1 TO INDICATE CONSOLE PACKAGE ACTIVE.

FOR A NON ACTIVE CONSOLE PACKAGE

- 1.) SET LOCATION 21 BIT0=1 TO INDICATE NOT TO USE PSEUDO SWITCH REGISTER, BUT TO USE HARDWARE SWITCHES.
- 2.) SET LOCATION 22 BIT3=0 TO INDICATE CONSOLE PACKAGE NOT ACTIVE.

9.4 CONTROL CHARACTERS

CONTROL CHARACTERS ARE USED TO GIVE THE OPERATOR THE ABILITY TO PERFORM THE FOLLOWING FUNCTIONS.
NOTE: THE PROGRAM WILL RESPOND TO THE CONTROL CHARACTER IN FIVE (5) SECONDS OR LESS.

CONTROL C

THIS WILL START THE LOADER THAT IS IN LOCATION 7600.

CONTROL R

THIS WILL RESTART THE PROGRAM AND REASK THE SWITCH REGISTER QUESTION AS DESCRIBED IN SECTION 9.6.

CONTROL E

THIS WILL CONTINUE THE PROGRAM FROM AN ERROR IF ALLOWED BY THE DIAGNOSTIC OR FROM A WAITING STATEMENT.

CONTROL L

THIS WILL SWITCH THE TERMINAL MESSAGES FROM THE DISPLAY TO A LINE PRINTER. TO RESTORE THE MESSAGES ON THE TERMINAL CONTROL L MUST BE TYPED AGAIN. IF NO PRINTER IS AVAILABLE AND CONTROL L IS TYPED THE RESULT WILL BE THAT THE CONSOLE PACKAGE WILL WAIT FOR CONTROL C OR R. THE CONTROL L WILL OUTPUT TO THE LINE PRINTER AND THE PROGRAM WILL ATTEMPT TO CONTINUE AS IF A CONTROL E WAS TYPED IN.

CONTROL D

THIS WILL ALLOW THE ABILITY TO CHANGE THE SWITCH REGISTER DURING PROGRAM OPERATION. TYPING THIS CHARACTER WILL RESULT IN AN INTERIGATION OF THE SWITCH REGISTER QUESTION AS DESCRIBED IN SECTION 9.6.

CONTROL S

THIS WILL STOP PROGRAM EXECUTION AND WAIT IN A LOOP FOR A CONTINUE. THE ONLY WAY TO CONTINUE WILL BE TO TYPE A CONTROL Q, R OR C. THIS IS A NONPRINTING CHARACTER.

CONTROL Q

THIS IS TO CONTINUE A PROGRAM AFTER A CONTROL S IS TYPED. THIS IS A NONPRINTING CHARACTER.

9.5 WAITING MESSAGE

THE WAITING MESSAGE IS USED TO ALLOW THE OPERATOR TIME TO MAKE A DECISION AS TO WHAT CONTROL CHARACTER

TO TYPE, THIS MESSAGE MAY APPEAR AT THE END
OF PASS MESSAGE IF THE HALT ON PASS BIT IS SET. THE CONTROL
CHARACTERS MAY NOW BE USED TO PERFORM THE NEEDED FUNCTION.

THE WAITING MESSAGE MAY BE PRINTED AFTER A ERROR MESSAGE
IF THE HALT ON ERROR BIT IS SET. HERE AGAIN THE CONTROL
CHARACTERS MAY BE USED.
THE WAITING MESSAGE MAY BE PRINTED IF OPERATOR INTERVENTION
IS REQUIRED.

9.6 SWITCH REGISTER MESSAGE

THIS MESSAGE IS USED TO SETUP THE PSEUDO SWITCH REGISTER
BEFORE PROGRAM EXECUTION TAKES PLACE. THE SWITCH REGISTER
IS SETUP WHEN THE FOURTH CHARACTER IS ENTERED OR A CARRIAGE
RETURN IS TYPED

SR=0000 4000

UNDER SCORING INDICATES OPERATOR RESPONSE

9.7 END OF PASS

THE NORMAL PROGRAM PASS COMPLETE AS DESCRIBED IN SECTION
4.4 IS USED.

9.8 ERRORS

THE STANDARD ERROR REPORTS AS DESCRIBED IN SECTION 5
OF THIS DOCUMENT WILL BE USED.

9.9 SWITCH REGISTER SETTINGS

THE STANDARD SWITCH SETTINGS AS DESCRIBED IN SECTION 4.5
OF THIS DOCUMENT WILL BE USED.

9.10 PARAMETER CONTROL WORDS

THE CONSOLE PACKAGE USES THE LOCATIONS 20 21 22 FOR THE
FOLLOWING PURPOSES.

LOCATION 20
PSEUDO SWITCH REGISTER

3637 IS THE LOCATION SET FOR THE NUMBER OF
FILLER CHARACTERS AFTER A CRLF SET TO FOUR (4)


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      /RK8E/RK8L DISK FORMATTER PROGRAM: MD=08-DHRKD=D
      /MAINDEC=08-DHRKD=D=0
      /
06740  DLSC=6740      /LOAD SECTOR COUNTER
06741  OSKP=6741      /SKIP UN TRANSFER DONE OR ERROR
06742  DCLR=6742      /CLEAR DISK CONTROL LOGIC
06743  DLAG=6743      /LOAD ADDRESS AND GO
06744  DLCA=6744      /LOAD CURRENT ADDRESS
06745  DRST=6745      /READ STATUS REGISTER
06746  DLOC=6746      /LOAD COMMAND REGISTER
06747  DMAN=6747      /LOAD MAINTENANCE
      /
04446  LDSC=JMS I    XXLDSC
04430  IOTCHN=JMS I  XCHANG
04431  LOUTRK=JMS I  XWTRK
04432  REDDSK=JMS I  XRDRK
04433  RECAL=JMS I   XRESTR
04434  RECEIV=JMS I  XWAIT
04435  KILBUF=JMS I  XKLBUF
04437  ERROR=JMS I   XERRO
04440  RSTAT=JMS I   XRUST
04444  LDADD=JMS I   XLDAO
04441  OSKSKP=JMS I  XSDKP
04442  LDCMD=JMS I   XLDCM
04443  LDCUR=JMS I   XLUCA
04445  CLRAL=JMS I   XCLOD
04447  PRNTER=JMS I  XPHN
04450  OCTEL=JMS I   XFROCT
04451  THUCT=JMS I   XTUCT
04456  TYPE=JMS I    XPHINT
04452  CRLF=JMS I    XCRLF
04424  APT8A=JMS I   XAPT8
04425  TIME=JMS I    XTIME
04427  TICK=JMS I    XTICK
04426  KAERRO=JMS I  XAERRO
      /
0000  *0
      /
0000  0304      304      /REV D
0001  5001      5001
0002  0002      0002
0003  0003      0003
      /
0010  *10
      /
0010  0000      AUTO10, 0
      /
0011  0000      AUTO11, 0
      /
0020  *20
      /
0020  0000      0000      /PSEUDO SWITCH REGISTER
0021  4000      4000      /CONTRUL WORD 1
0022  0000      0000      /CONTRUL WORD 2

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0023  0000      0000      /RESERVED
0024  1125      APT8, APT8
0025  1557      XTIME, KTIME
0026  1600      XAERRO, AERRO
0027  1530      XTICK, KTICK
0030  1463      XCHANG, CHANG
0031  0600      XWTRK, WTRK
0032  1000      XRDRK, REDTRK
0033  1400      XRESTR, RESTOR
0034  1327      XWAIT, WAIT
0035  0752      XKLBUF, KLBUF
0036  1312      XPHINT, PRINT
0037  0436      XERRO, ERRO
0040  0671      XRUST, ROST
0041  0740      XSDKP, SDKP
0042  0720      XLDCM, LDCM
0043  0700      XLUCA, LOCA
0044  0711      XLDAO, LOAD
0045  0745      XCLOD, CLOR
0046  0733      XXLDSC, XLDSK
0047  1252      XPHN, PRN
0050  1227      XFROCT, FROCT
0051  1200      XTUCT, TOCT
0052  1215      XCRLF, UPONE
0053  2201      XLOTRK, LOTRK
0054  2200      XMITRK, HITRK
0055  2200      BGNBUF, WRKBUF
0056  0000      AMOUNT, 0
0057  0000      SWITCH, 0
0060  0003      K0003, 0003
0061  0004      K4, 4
0062  0007      K0007, 0007
0063  0040      K0040, 0040
0064  7465      M313, -313
0065  0277      K0277, 0277
0066  0200      K0200, 0200
0067  0260      K0260, 0260
0070  4000      K4000, 4000
0071  7735      K7735, 7735
0072  7760      K7760, 7760
0073  0400      K0400, 400
0074  0037      K0037, 0037
0075  6201      KCDF, CDF
0076  7774      M4, -4
0077  7770      M10, -10
0100  0000      DRIVNO, 0
0101  0000      CHAR, 0
0102  0000      LOWAD, 0
0103  0000      HIGHAD, 0
0104  0000      TRKCNT, 0
0105  0000      DSKCNT, 0
0106  0000      SBCNT1, 0
0107  0000      STCNT1, 0
0110  0000      STCNT2, 0
0111  0000      STCNT3, 0

```

```

0112 0000 TCNTR1, 0
0113 0000 TCNTR2, 0
0114 0000 TCNTR3, 0
0115 0000 TCNTR4, 0
0116 0000 TCNTR5, 0
/
0117 0000 GOREG2, 0
0120 0000 EX8IT, 0
0121 0000 CMREG, 0
0122 0000 STREG, 0
0123 0000 DAREG, 0
0124 0000 CAREG, 0
0125 0000 ADREG, 0
0126 0000 DTREG, 0
0127 0263 BGNTRST, FRMDSK
0130 0000 HOMEHA, 0
0131 0000 DATCNT, 0
0132 7776 CLKCNT, -2
/
0133 1623 XMOVE, MOVE
0134 0000 LOC8ED, 0
0135 0424 XEND, ENDTST
0136 0000 SOFT, 0
0137 0140 ADPOT1, DSK8A
0140 0000 DSK8A, 0
0141 0000 DSK1A, 0
0142 0000 DSK2A, 0
0143 0000 DSK3A, 0
0144 0000 DSK4A, 0
0145 0000 DSK5A, 0
0146 0000 DSK6A, 0
0147 0000 DSK7A, 0
0150 0151 ADPOT2, DSK8B
0151 0000 DSK8B, 0
0152 0000 DSK1B, 0
0153 0000 DSK2B, 0
0154 0000 DSK3B, 0
0155 0000 DSK4B, 0
0156 0000 DSK5B, 0
0157 0000 DSK6B, 0
0160 0000 DSK7B, 0
0161 0000 PCOUNT, 0
/
0200 *200
/
0200 6224 BGN, RIF
0201 3130 DCA HOMEHA
0202 1130 TAD HOMEHA
0203 1075 TAD KCDF
0204 3205 DCA ,+1
0205 7402 MLT
/NOH TEST FOR APT SYSTEM
/IF ON APT TERMINAL MESSAGES ARE SKIP
/TO AVOID TIMING PROBLEMS WITH THE SYSTEM
0206 4424 APT8A

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/USED ONLY IF ON APT

/MAKE HOMEDF

/MAKE DF=IF

/TEST FOR APT SYSTEM

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0207 4777 JMS XC8PSW
0210 4430 IOTCHN
0211 4452 CRLF
0212 4452 CRLF
0213 4447 PRNTER
0214 2045 MES1
0215 4452 CRLF
0216 4447 PRNTER
0217 2066 MES2
0220 1077 ALLAGN, TAD M10
0221 3107 DCA STCNT1
0222 3134 DCA LOC8ED
0223 3110 DCA STCNT2
0224 4452 SAMAGN, CRLF
0225 4447 PRNTER
0226 2117 MES3
0227 1110 TAD STCNT2
0230 1067 TAD K0260
0231 4436 TYPE
0232 1065 GUES1, TAD K0277
0233 4436 TYPE
0234 1137 TAD ADPOT1
0235 1110 TAD STCNT2
0236 3111 DCA STCNT3
0237 4434 RECEIV
0240 5244 JMP NOTDSK
0241 5232 JMP GUES1
0242 2134 WASDSK, ISZ LOC8ED
0243 7340 CLA CLL CMA
0244 3511 NOTDSK, DCA I STCNT3
0245 2110 ISZ STCNT2
0246 2107 ISZ STCNT1
0247 5224 JMP SAMAGN
/
0250 4452 DONE, CRLF
0251 4447 PRNTER
0252 2126 MES4
0253 4434 RECEIV
0254 5220 JMP ALLAGN
0255 5250 JMP DONE
0256 1134 TAD LOC8ED
0257 7041 CIA
0260 7450 SNA
0261 5200 JMP BGN
0262 3134 DCA LOC8ED
/
0263 4533 FRMDSK, JMS I XMOVE
0264 1134 TAD LOC8ED
0265 3056 DCA AMOUNT
0266 1056 TAD AMOUNT

```

/GET SR=

/CHANGE DEVICE TO SWR3=8

/PRINT "RK8E/RK8L DISK FORMATTER PROGRAM"

/MESSAGE 1 POINTER

/PRINT "FOR ALL QUESTIONS"

/MESSAGE POINTER 2

/COUNTER FOR AMOUNT OF DISKS

/PRINT "FORMAT DISK ?"

/MESSAGE POINTER 3

/TYPE DISK NUMBER

/TYPE ?

/WAIT FOR CHARACTER

/NO NOT THIS DISK

/NEITHER YES OR NO

/YES, WAS CLEAR DISK POINTER

/UPDATE POINTER

/COUNT DISKS

/ASK ABOUT NEXT

/PRINT "ARE YOU SURE ?"

/MESSAGE POINTER 4

/WAIT FOR CHARACTER

/NO, START ALL OVER

/NEITHER TYPE ?

/ANY DISKS

/NO, OPERATOR ERROR

/YES, AMOUNT LOCATED

/FIRST RECALIBRATE AND FORMAT IN WRITE ALL MODE

/ALL DISK DRIVES SELECTED BY OPERATOR,, MAKE THE FIRST

/TWO WORDS OF EVERY DISK SECTOR EQUAL TO THE

/ABSOLUTE DISK ADDRESS.

/MOVE DISK POINTERS

```

0267 3105      OCA   DSKCNT
0270 3115      OCA   TCNTR4
0271 1150      TAD   ADPOT2
0272 3116      OCA   TCNTR5
0273 1516      TAD   I   TCNTR5
0274 7640      SZA   CLA
0275 5302      JMP     FOMHAT
0276 2116      NEXFRM, ISZ   TCNTR5
0277 2115      ISZ   TCNTR4
0300 5273      JMP     .-5
0301 7402      HLT

/ FOMHAT, TAD   TCNTR4
0302 1115      AND   K0003
0303 0060
0304 7104      CLL   RAL
0305 3100      OCA   DRIVNO
0306 1115      TAD   TCNTR4
0307 0061      AND   K4
0310 7640      SZA   CLA
0311 1066      TAD   K0200
0312 3120      OCA   EXBIT
0313 4433      RECAL
0314 5335      JMP     RENEX1
0315 3102      OCA   LOWAD
0316 3103      OCA   HIGHAD
0317 1064      TAD   M313
0320 3104      OCA   TRKCNT

/
/ WRDTSK, TICK
0321 4427      =4
0322 7774      LODTRK
0323 4431      JMP     RENEX1
0324 5335
0325 7300      CLA   CLL
0326 1102      TAD   LOWAD
0327 1063      TAD   K0040
0330 3102      OCA   LOWAD
0331 7630      SZA   CLA
0332 2103      ISZ   HIGHAD
0333 2104      ISZ   TRKCNT
0334 5321      JMP     WRDTSK
0335 2105      RENEX1, ISZ   DSKCNT
0336 5276      JMP     NEXFRM

/ ROUTINE TO CHECK ADDRESSING INFORMATION ON THE DISK,
/ THE FIRST TWO WORDS OF EVERY SECTOR SHOULD EQUAL
/ THE ABSOLUTE DISK ADDRESS. ALL OTHER DATA IS
/ NOT CHECKED.
0337 1056      CHKDSK, TAD   AMOUNT
0340 3105      OCA   DSKCNT
0341 3115      OCA   TCNTR4
0342 1150      TAD   ADPOT2
0343 3116      OCA   TCNTR5
0344 1516      TAD   I   TCNTR5

```

/AMOUNT OF DISKS

/SOFTWARE INFORMATION

```

0345 7640      SZA   CLA
0346 5353      JMP     CHKDAT
0347 2116      NEXCHK, ISZ   TCNTR5
0350 2115      ISZ   TCNTR4
0351 5344      JMP     .-5
0352 7402      HLT

/ CHKDAT, TAD   TCNTR4
0353 1115      AND   K0003
0354 0060
0355 7104      CLL   RAL
0356 3100      OCA   DRIVNO
0357 1115      TAD   TCNTR4
0360 0061      AND   K4
0361 7640      SZA   CLA
0362 1066      TAD   K0200
0363 3120      OCA   EXBIT
0364 4433      RECAL
0365 5776      JMP     RENEX2
0366 3102      OCA   LOWAD
0367 3103      OCA   HIGHAD
0370 1064      TAD   M313
0371 3104      OCA   TRKCNT
0372 5775      JMP     CHECK

/
0375 0400
0376 0414
0377 3456
0400 0400      PAGE

/ CHECK, TICK
0400 4427      =4
0401 7774      REDDSK
0402 4432      JMP     RENEX2
0403 5214
0404 7300      CLA   CLL
0405 1102      TAD   LOWAD
0406 1063      TAD   K0040
0407 3102      OCA   LOWAD
0410 7630      SZA   CLA
0411 2103      ISZ   HIGHAD
0412 2104      ISZ   TRKCNT
0413 5200      JMP     CHECK
0414 2105      RENEX2, ISZ   DSKCNT
0415 5777      JMP     NEXCHK

/
0416 1022      TAD   22
0417 0070      AND   K4000
0420 7630      SZA   CLA
0421 5224      JMP     ENDTST
0422 2161      ISZ   PCOUNT
0423 5776      JMP     FRMDSK
0424 4452      ENDTST, CRLF
0425 4447      PRNTER
0426 2021      TEXEND
0427 4452      CRLF

```

/CHECK THIS DISK
/CHECK THIS ONE
/UPDATE FOR NEXT DISK

/WHAT HAPPENED?????

/MASK OUT
/MAKE DRIVE NUMBER/SET EXTENDED DRIVE BIT
/RECALIBRATE
/TRY NEXT DRIVE
/SETUP STARTING DISK ADDRESS
/AMOUNT OF TRACKS TO DO/TIMING FOR APT IF NEEDED,
/SKIPPED IF NOT REQUIRED,
/READ AND CHECK ONE CYLINDER
/TO NEXT DISK/UPDATE TO NEXT CYLINDER
/TIME TO SET EXTENDED BIT
/YES, SET IT
/UPDATE CYLINDER COUNTER
/CHECK NEXT ONE
/UPDATE DISK COUNTER
/CHECK NEXT/TEST FOR APT
/ARE WE?
/NO, NORMAL RUN
/INCREMENT PASS COUNT
/LOOP PROGRAM
/PRINT "PASS COMPLETE"

```

0430 4447      PRNTER      /PRINT "TRY SAME SEQUENCE"
0431 2135      MESS
0432 4434      RECEIV      /WAIT FOR INPUT FROM OPERATOR
0433 5775*     JMP ALLAGN    /NO, ASK AGAIN
0434 5227      JMP .+3
0435 5776*     JMP FRMDSK    /TRY SAME SEQUENCE

/
/SUBROUTINE FOR "ERRORS," SCOPE LOOPS, AND
/ERROR TYPEOUTS.
/
0436 0000      ERNO, 0
0437 7301      CLA CLL IAC
0440 1236      TAD ERNO      /GET PC STORED
0441 3344      DCA RETRN1    /STORE FOR RETURN
0442 4426      KAERRO
0443 4452      CRLF         /NOTIFY APT OF ERROR IS NEEDED
0444 4452      CRLF
0445 1636      TAD I ERNO    /GET TEXT POINTER
0446 0062      AND K0007     /MASK 9-11
0447 1352      TAD HEDTAD    /MAKE ERROR HEADER TAD
0450 3251      DCA .+1
0451 7402      HLT          /MODIFIED HEADER TAD
0452 3254      DCA .+2
0453 4447      PRNTER      /MODIFIED HEADER POINTER
0454 7402      HLT
0455 4452      CRLF
0456 4447      PRNTER      /PRINT PC:
0457 1642      TEXPC
0460 1236      TAD ERNO      /GET PC POINTER
0461 4450      OCTEL        /PRINT PC STORED
0462 1636      TAD I ERNO    /GET TEXT POINTER
0463 7104      CLL RAL
0464 7420      SNL
0465 5274      JMP NTGD      /NOT GD: REGISTER
0466 3236      DCA ERNO
0467 4447      PRNTER      /PRINT GD:
0470 1644      TEXGD
0471 1117      TAD GOME62
0472 4450      OCTEL        /PRINT FOUR OCTAL
0473 7610      SKP CLA
0474 3236      DCA ERNO
0475 4447      PRNTER
0476 1646      TEXEX
0477 1120      TAD EXBIT
0500 7640      SZA CLA
0501 7001      IAC
0502 4450      OCTEL
0503 1345      TAD XTEXT
0504 3350      DCA PCNTR2
0505 1346      TAD XREG
0506 3410      UCA AUTO10
0507 1357      TAD K7771
0510 3347      DCA PCNTR1    /COUNTER FOR # OF HEADS
0511 7344      CLA CLL CMA RAL

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0512 3351      DCA PCNTR3
0513 1236      STRAUT, TAD ERNO /GET TEXT POINTER
0514 7500      SNA
0515 5336      JMP NOTEX      /NOT THIS ONE
0516 7104      CLL RAL
0517 3236      DCA ERNO
0520 1350      TAD PCNTR2
0521 2350      ISZ PCNTR2
0522 2350      ISZ PCNTR2
0523 3325      DCA .+2
0524 4447      PRNTER
0525 7402      HLT
0526 1410      TAD I AUTO10
0527 4450      OCTEL
0530 2351      ISZ PCNTR3
0531 7610      SKP CLA
0532 4452      CRLF
0533 2347      AGAIN, ISZ PCNTR1
0534 5313      JMP STRAUT
0535 5744      JMP I RETRN1
0536 7104      NOTEX, CLL RAL
0537 3236      DCA ERNO
0540 2350      ISZ PCNTR2
0541 2350      ISZ PCNTR2
0542 2010      ISZ AUTO10
0543 5333      JMP AGAIN

/
0544 0000      RETRN, 0
0545 1650      XTEXT, TEXCM
0546 0120      XREG, EXBIT
0547 0000      PCNTR1, 0
0550 0000      PCNTR2, 0
0551 0000      PCNTR3, 0
0552 1353      HEDTAD, TAD HEDLST
0553 1664      HEDLST, ERTX1
0554 1675      ERTX2
0555 1705      ERTX3
0556 1717      ERTX4
0557 7771      K7771, 7771
/

0575 0220      PAGE
0576 0263
0577 0347
0600 0000
/ROUTINE TO FORMAT CYLINDER
/MAKE FIRST TWO WORDS OF EVERY SECTOR
/EQUAL TO DISK ADDRESS.
/
0600 0000      WRTTRK, 0
0601 7330      CLA CLL CML RAR
0602 3117      DCA GOREG2
0603 4435      KILBUF
0604 1071      TAD K7735
0605 3112      DCA TCNTR1
/SETUP COMPARE REGISTER
/CLEAR BUFFER
/AMOUNT OF SECTORS TO DO
/SETUP COUNTER

```

```

0606 3113      DCA   TCNTR2      /STARTING WITH 0
0607 1072      TAD   K7760        /STOPPER
0610 3114      DCA   TCNTR3      /SECTOR COUNTER POINTER STOP
0611 1113      LODR1, TAD   TCNTR2
0612 0074      AND   K0037        /MASK SECTOR BITS
0613 1102      TAD   LOWAD        /ADD IN CYLINDER
0614 3453      DCA I  XLOTRK      /SETUP TRACK WORD IN BUFFER
0615 1120      TAD   EXBIT        /ADD IN EXTENDED BIT
0616 1103      TAD   HIGHAD       /
0617 1100      TAD   DRIVNO       /ADD IN DRIVE NUMBER
0620 3454      DCA I  XMITRK      /SETUP TRACK WORD IN BUFFER
0621 1454      TAD I  XMITRK
0622 0270      AND   K7577
0623 1130      TAD   MOMEMA       /CURRENT FIELD
0624 1267      TAD   K5000        /FUNCTION WRITE ALL
0625 4442      LDCHD             /LOAD COMMAND
0626 1120      TAD   EXBIT
0627 4446      LDSC             /LOAD EXTENDED DRIVE BIT
0630 7200      CLA              /CLEAR EXTENDED DRIVE BIT
0631 1055      TAD   BGNBUF
0632 4443      LOCUR             /LOAD CURRENT ADDRESS
0633 1453      TAD I  XLOTRK
0634 4444      LDADD            /LOAD TRACK AND GO
0635 4441      DSKSKP          /SKIP ON FLAG
0636 5235      JMP   .-1         /WAIT FOR FLAG
0637 4440      ROSTAT          /READ STATUS
0640 1070      TAD   K4000
0641 7640      SZA CLA          /WAS STATUS 0?
0642 5254      JMP   LOUER       /ERROR, STATUS ON WRITE ALL
0643 2113      ISZ   TCNTR2
0644 2114      ISZ   TCNTR3      /COUNT FIRST REVOLUTION
0645 7610      SKP CLA          /STILL IN FIRST REV.
0646 3113      DCA   TCNTR2      /SETUP FOR SECTOR "1"
0647 2113      ISZ   TCNTR2
0650 2112      ISZ   TCNTR1
0651 5211      JMP   LODR1       /UPDATE SECTOR COUNTER
0652 2200      ISZ   WRITRK      /TRY NEXT SECTOR
0653 5600      JMP I  WRITRK
0654 4437      LODER, ERROR      /THIS CYLINDER DONE
0655 3602      3602            /ERROR, STATUS
                                /TEXT POINTER
/
0656 4433      RECAL            /CLEAR CONTROL AND DRIVE
0657 5600      JMP I  WRITRK     /TO NEXT DISK
0660 4452      CRLF
0661 4447      PRNTER          /PRINT "TRY SAME AGAIN"
0662 1734      ERMES1
0663 4434      RECEIV
0664 5252      JMP   LOUER=2     /WAIT FOR YES OR NO
0665 5260      JMP   .-3         /WAS A NO TRY SAME CYLINDER
0666 5201      JMP   WRITRK +1   /WAS NEITHER ASK AGAIN
0667 5000      K5000, 5000      /YES, TRY NEXT
0670 7577      K7577, 7577
/
/
/SUBROUTINE TO READ STATUS REGISTER

```

```

/
0671 0000      ROST, 0
0672 6745      IOT5, DRST      /READ STATUS IOT
0673 7410      SKP
0674 4777*     ERMLT5, JMS      XCERR /SKIP TRAP ERROR.
0675 3122      DCA   STNEG      /SAVE RESULTS
0676 1122      TAD   STNEG
0677 5671      JMP I  ROST      /EXIT
/
/SUBROUTINE TO LOAD CURRENT ADDRESS REGISTER
/
0700 0000      LDCA, 0
0701 3125      DCA   ADNEG      /SAVE IN ADDRESS
0702 1125      TAD   ADNEG
0703 3124      DCA   CAHEG      /SETUP INITIAL CURRENT ADDRESS
0704 1125      TAD   ADNEG
0705 6744      IOT4, ULCA      /LOAD CURRENT ADDRESS IOT
0706 5700      JMP I  LDCA      /EXIT
0707 4777*     ERMLT4, JMS      XCERR /SKIP TRAP ERROR.
0710 5307      JMP   .-1
/
/
/SUBROUTINE TO LOAD TRACK ADDRESS REGISTER
/
0711 0000      LOAD, 0
0712 3123      DCA   DAHEG      /SAVE OUTBOUND DATA
0713 1123      TAD   DAHEG
0714 6743      IOT3, DLAG      /LOAD DISK ADDRESS REGISTER
0715 5711      JMP I  LOAD      /EXIT
0716 4777*     ERMLT3, JMS      XCERR /SKIP TRAP ERROR.
0717 5316      JMP   .-1
/
/
/SUBROUTINE TO LOAD COMMAND REGISTER
/
0720 0000      LDCH, 0
0721 3121      DCA   CMREG      /SAVE OUTBOUND DATA
0722 3776*     DCA   INMODE
0723 4775*     JMS   XCCKP      /CHECK FOR CONTROL CHARACTERS.
0724 7200      CLA
0725 7200      CLA
0726 1121      TAD   CMREG
0727 6746      IOT6, OLDC      /LOAD COMMAND REGISTER
0730 5720      JMP I  LDCH      /EXIT
0731 4777*     ERMLT6, JMS      XCERR /SKIP TRAP ERROR.
0732 5331      JMP   .-1
/
/
/SUBROUTINE ISSUE "OLSC"
0733 0000      XLSC, 0
0734 6740      IOT0, OLSC
0735 5733      JMP I  XLSC
0736 4777*     ERMLT0, JMS      XCERR
0737 5336      JMP   .-1

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      /SUBROUTINE TO ISSUE "DSKP" DISK SKIP IOT
      /
0740 0000      SDKP, 0
0741 0741      IOT1, DSKP      /DISK SKIP IOT
0742 7410      SKP      /DID NOT SKIP
0743 2340      ISZ      SDKP
0744 5740      JMP I      SDKP      /EXIT
      /
      /SUBROUTINE TO ISSUE "DCLR" CLEAR IOT
      /
0745 0000      CLDR, 0
0746 0742      IOT2, DCLR      /DCLR "CLEAR IOT"
0747 5745      JMP I      CLDR      /EXIT
0750 4777      ERHLT2, JMS      XCERR      /SKIP TRAP ERROR.
0751 5350      JMP      .-1
      /
      /ROUTINE TO ZERO WORK BUFFER
      /
0752 0000      KLBUF, 0
0753 7340      CLA CLL CMA
0754 1055      TAD      BGNBUF      /START OF BUFFER =1
0755 3010      DCA      AUTO10      /SETUP AUTO INDEX
0756 1364      TAD      K7400
0757 3131      DCA      UATCNT      /SETUP COUNTER
0760 3410      DCA I      AUTO10      /CLEAR BUFFER
0761 2131      ISZ      UATCNT      /UPDATE COUNTER
0762 5360      JMP      .-2      /NOT ALL CLEARED YET
0763 5752      JMP I      KLBUF      /BUFFER CLEARED
0764 7400      K7400, 7400
      /
0775 3641
0776 3676
0777 4007
      1000
      PAGE
      /
      /ROUTINE TO READ AND CHECK A CYLINDER
      /
1000 0000      REDTRK, 0
1001 1071      TAD      K7735
1002 3112      DCA      TCNTR1      /AMOUNT OF SECTORS TO DO
1003 3113      DCA      TCNTR2      /STARTING WITH 0
1004 1072      TAD      K7760
1005 3114      DCA      TCNTR3
1006 4435      KILBUF      /CLEAR BUFFER
1007 7340      CHKR1, CLA CLL CMA
1010 3136      DCA      SOFT      /SETUP SOFT ERROR FLAG
1011 1055      TAD      BGNBUF
1012 4443      LDCUR      /LOAD CURRENT ADDRESS
1013 1103      TAD      MIGHAD      /EXTENDED CYLINDER BIT
1014 1100      TAD      URIVNO      /CURRENT DRIVE
1015 1130      TAD      HOMEMA      /CURRENT FIELD
1016 4442      LDCMD      /LOAD COMMAND
1017 1120      TAD      EXBIT      /LOAD EXTENDED DRIVE BIT
1020 4446      LDBC
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1021 7200      CLA      /CLEAR EXTENDED DRIVE BIT
1022 1113      TAD      TCNTR2
1023 0074      AND      K0037      /MASK SECTOR BITS OFF
1024 1102      TAD      LOWAD      /ADD IN OTHER DISK ADDRESS
1025 4444      LDADD      /LOAD AND GO
1026 4441      DSKSKP      /DISK SKIP IOT
1027 5226      JMP      .-1      /WAIT FOR FLAG
1030 4440      RDSTAT      /READ STATUS
1031 1070      TAD      K4000      /ADD IN FUDGE FACTOR
1032 7650      SNA CLA      /SKIP IF ERROR
1033 5241      JMP      STAOK      /STATUS O.K.
1034 1122      TAD      STREG      /GET STATUS READ
1035 0777      AND      K0010
1036 7650      SNA CLA      /WAS IT A CRC
1037 5306      JMP      STAER      /NO, JUST A HARD ERROR
1040 3136      DCA      SOFT      /CLEAR SOFT ERROR FLAG
1041 1121      STAOK, TAD      CMREG      /GET LAST COMMAND
1042 0062      AND      K0007
1043 1120      TAD      EXBIT      /ADD EXTENDED DRIVE BIT
1044 7041      CIA
1045 1454      TAD I      XMITRK      /GET WORD READ FROM DISK
1046 7650      SNA CLA      /SKIP IF ERROR
1047 5256      JMP      FRSTOK      /FIRST WORD O.K.
1050 1454      TAD I      XMITRK      /GET WORD
1051 3126      DCA      DTNEG      /SETUP ERROR PRINTER
1052 1121      TAD      CMREG
1053 0062      AND      K0007
1054 3117      DCA      GDNEG2      /SETUP GOOD FOR PRINTER
1055 5303      JMP      DATER      /NO, DATA ERROR
1056 1453      FRSTOK, TAD I      XLOTRK      /GET WORD READ
1057 7041      CIA
1060 1123      TAD      DANEG      /COMPARE TO GOOD
1061 7650      SNA CLA      /SKIP IF ERROR
1062 5271      JMP      DATOK      /WORD O.K.
1063 2125      ISZ      ADNEG      /SETUP ERROR PRINTER
1064 1123      TAD      DANEG
1065 3117      DCA      GDREG2      /SETUP GOOD WORD FOR PRINTER
1066 1453      TAD I      XLOTRK      /GET WORD READ
1067 3126      DCA      DTNEG      /SETUP FOR PRINTER
1070 5303      JMP      DATER      /DATA ERROR
1071 1136      DATOK, TAD      SOFT      /GET SOFT ERROR FLAG
1072 7650      SNA CLA      /WAS IT CLEAR
1073 5306      JMP      STAER      /YES, STATUS ERROR
1074 1113      TAD      TCNTR2
1075 1000      TAD      K0003      /ADVANCE 3 SECTORS
1076 3113      DCA      TCNTR2
1077 2114      ISZ      TCNTR3
1080 5207      JMP      CHKR1      /MORE TO FORMAT
1081 2200      ISZ      REDTRK
1082 5600      JMP I      REDTRK      /EXIT, O.K.
1083 1776      DATER, TAD      K7741
1084 3313      DCA      TCHKT      /SETUP TEXT PRINTER
1085 5312      JMP      CHKR      /ERROR
1086 1775      STAER, TAD      K3600
1087 3313      DCA      TCHKT      /SETUP TEXT PRINTER
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1110 7330      CLA CLL CML RAR
1111 3117      DCA GOMEG2      /SETUP GOOD STATUS PRINTEN
1112 4437      CHKER, ERROR      /ERRUR, READ DATA
1113 0000      TCHKT, 0      /MODIFIED TEXT POINTER
1114 4433      RECAL      /CLEAR CONTROL AND DRIVE
1115 5600      JMP I REUTRK      /TO NEXT DISK
1116 4452      CRLF
1117 4447      PRNTER      /PRINT "TRY SAME AGAIN"
1120 2000      ERMES3
1121 4434      RECEIV
1122 5301      JMP DATER =2      /CHECK NEXT
1123 5316      JMP ,=3      /RE-PRINT
1124 5201      JMP REUTRK +1      /TRY SAME AGAIN

/
/THIS ROUTINE WILL TEST FOR APT AND NOP CONSOLE
/PACKAGE IF NEED BE
/
1125 0000      APT0, 0
1126 1022      TAD 22
1127 7700      SMA CLA
1130 5725      JMP I APT0
1131 1022      TAD 22
1132 0373      AND K7377      /ON APT, NOP CONSOLE PACKAGE
1133 3022      DCA 22
1134 1022      TAD 22
1135 0062      AND K0007      /ISOLATE DRIVE NUMBER OR
                                /NUMBER OF DRIVES TO BE DONE

1136 3107      DCA STCNT1
1137 1022      TAD 22
1140 0774      AND K0100
1141 7650      SMA CLA
1142 5353      JMP MULDSK      /SINGLE DRIVE TESTING
1143 1137      TAD ADPOT1      /NO, SEVERAL TO DO
1144 1107      TAD STCNT1      /GET DISK POINTER
1145 3107      DCA STCNT1      /ESTABLISH DRIVE TO DO
1146 7340      CLL CLA CMA      /-1
1147 3507      DCA I STCNT1
1150 7340      CLL CLA CMA      /ONE DISK TO DO
1151 3134      DCA LOC8ED
1152 5527      JMP I 0GNTST
1153 1107      MULDSK, TAD STCNT1      /UNIVE TO BE DONE
1154 7040      CMA
1155 3107      DCA STCNT1
1156 1137      TAD ADPOT1      /GET DISK POINTER
1157 1110      TAD STCNT2      /ESTABLISH DRIVE TO BE DONE
1160 3111      DCA STCNT3
1161 2134      ISZ LOC8ED
1162 7340      CLL CLA CMA
1163 3511      DCA I STCNT3      /DO THIS DRIVE
1164 2110      ISZ STCNT2
1165 2107      ISZ STCNT1
1166 5356      JMP MULDSK+3      /MORE TO DO
1167 1134      TAD LOC8ED
1170 7041      CIA
1171 3134      DCA LOC8ED      /NUMBER TO BE DONE

```

```

1172 5527      JMP I 0GNTST
1173 7377      K7377, 7377
1174 1556
1175 1326
1176 1325
1177 1324
1178 1200

PAGE
/
/SUBROUTINE TO PRINT TWO OCTAL
/
1200 0000      TOCT, 0
1201 3106      DCA SBCNT1      /SAVE AC
1202 1106      TAD SBCNT1
1203 7010      RAR
1204 7012      RTR
1205 0062      AND K0007
1206 1067      TAD K0260
1207 4436      TYPE      /PRINT FIRST BYTE
1210 1106      TAD SBCNT1
1211 0062      AND K0007
1212 1067      TAD K0260
1213 4436      TYPE      /PRINT SECOND BIT
1214 5600      JMP I TOCT      /EXIT

/
/
/ROUTINE TO DO CRLF
/
1215 0000      UPONE, 0
1216 7300      CLA CLL
1217 1225      TAD K0215
1220 4436      TYPE
1221 1226      TAD K0212
1222 4436      TYPE
1223 4436      TYPE      /TYPE ONE NULL
1224 5615      JMP I UPONE

1225 0215      K0215, 0215
1226 0212      K0212, 0212

/ROUTINE TO PRINT FOUR OCTAL
/
1227 0000      FROCT, 0
1230 7006      RTL
1231 7006      RTL
1232 3215      DCA UPONE
1233 1076      TAD M4
1234 3200      DCA TOCT
1235 1215      TAD UPONE
1236 0062      AND K0007
1237 1067      TAD K0260
1240 4436      TYPE
1241 1215      TAD UPONE
1242 7006      RTL
1243 7004      RAL

```

```

1244 3215      DCA  UPONE
1245 2200      ISZ  TUCT
1246 5235      JMP  ,=11
1247 1321      TAD  K0240
1250 4436      TYPE
1251 5627      JMP I  FROCT

```

```

/
/SUBROUTINE TO PRINT TEXT
/

```

```

1252 0000      PRN,  0
1253 7300      CLA CLL
1254 1652      TAD I  PRN          /GET POINTER
1255 2252      ISZ  PRN
1256 3227      DCA  FROCT
1257 1627      TAD I  FROCT
1260 0322      AND  K7700
1261 7450      SNA
1262 5306      JMP  EXIT
1263 7500      SMA
1264 7020      CML
1265 7001      IAC
1266 7012      RTH
1267 7012      RTR
1270 7012      RTR
1271 4436      TYPE
1272 1627      TAD I  FROCT
1273 0323      AND  K0077
1274 7450      SNA
1275 5306      JMP  EXIT
1276 1311      TAD  K3740
1277 7500      SMA
1280 1310      TAD  K4100
1281 1321      TAD  K0240
1282 4436      TYPE
1283 2227      ISZ  FROCT
1284 7300      CLA CLL
1285 5257      JMP  PRN+5
1286 7300      EXIT,  CLA CLL
1287 5652      JMP I  PRN

```

```

/
K4100, 4100
K3740, 3740
/

```

```

/ROUTINE TO TYPE
/

```

```

1312 0000      PRINT, 0
1313 0046      TLS
1314 0041      TSF
1315 5314      JMP  ,=1
1316 0042      TCF
1317 7200      CLA
1320 5712      JMP I  PRINT
1321 0240      K0240, 0240
1322 7700      K7700, 7700

```

```

1323 0077      K0077, 0077
1324 0010      K0010, 10
1325 7741      K7741, 7741
1326 3600      K3600, 3600

```

```

/ROUTINE TO WAIT FOR KEY FROM OPERATOR
/

```

```

1327 0000      WAIT,  0
1330 7300      CLA CLL
1331 6032      KCC
1332 6031      KSF
1333 5332      JMP  ,=1
1334 6036      KRB
1335 6046      TLS
1336 6041      TSF
1337 5336      JMP  ,=1
1340 0370      AND  K0177
1341 1066      TAD  K0200
1342 3101      DCA  CHAR
1343 1101      TAD  CHAR
1344 3777*     DCA  C0CHAR
1345 2776*     ISZ  INMODE
1346 4775*     JMS  XC0CNT          /CHECK FOR CONTROL CHARACTERS,
1347 7200      CLA
1350 7200      CLA
1351 3776*     DCA  INMODE
1352 6032      KCC
1353 6042      TCF
1354 1101      TAO  CHAR
1355 7041      CIA
1356 1371      TAD  K0316
1357 7650      SNA CLA
1360 5727      JMP I  WAIT          /WAS IT A NO
1361 2327      ISZ  WAIT          /YES
1362 1101      TAD  CHAR          /UPDATE RETURN POINTER
1363 7041      CIA
1364 1372      TAD  K0331
1365 7650      SNA CLA
1366 2327      ISZ  WAIT          /WAS IT A YES
1367 5727      JMP I  WAIT          /WAS A YES
1370 0177      K0177, 0177          /WAS NEITHER
1371 0316      K0316, 0316
1372 0331      K0331, 0331
/

```

```

1375 3200
1376 3676
1377 3675
1400 1400

```

```

PAGE
/
/

```

```

/ROUTINE TO RECALIBRATE SELECTED DRIVE
/

```

```

1400 0000      RESTOR, 0
1401 7301      CLA CLL IAC          /ENABLE CLEAR CONTROL
1402 4445      CLRALL              /CLEAR CONTROL
1403 1100      TAD  DRVNO          /CURRENT DRIVE

```

```

1404 1130      TAD      HOMEPA      /CURRENT FIELD
1405 4442      LOCMD      /LOAD COMMAND
1406 1120      TAD      EXBIT
1407 4446      LOBC
1410 7330      CLA CLL CML RAR      /LOAD EXTENDED DRIVE BIT
1411 3117      DCA      GDMG2      /MAYBE EXPECTED STATUS
1412 7326      CLA CLL CML RTL      /SETUP COMPARE REGISTER
1413 4445      CLRALL      /ENABLE RECALIBRATE BIT
1414 4441      DSKSKP      /"RECALIBRATE"
1415 5214      JMP      *-1      /DISK SKIP IOT
1416 4440      ROSTAT      /WAIT FOR FIRST DONE FLAG
1417 1327      TAD      K2000      /READ STATUS
1420 7450      SNA
1421 5225      JMP      RESTA      /WAS IT BUSY AND DONE
1422 1327      TAD      K2000      /YES, THEN ITS D.K.
1423 7640      SZA CLA      /NO, THEN IT MUST BE JUST DONE
1424 5243      JMP      RESTER      /WAS IT JUST DONE
1425 4445      RESTA, CLRALL      /NO, ERROR
1426 1066      TAD      K0200      /CLEAR STATUS
1427 1121      TAD      CMREG      /ENABLE SET SECOND DONE FLAG
1430 4442      LOCMD      /ORIGINAL COMMAND
1431 4441      DSKSKP      /LOAD COMMAND
1432 5231      JMP      *-1      /DISK SKIP IOT
1433 4440      ROSTAT      /WAIT FOR SECOND DONE
1434 1070      TAD      K4000      /READ STATUS
1435 7640      SZA CLA      /WAS IT ONLY DONE FLAG
1436 5243      JMP      RESTER      /NO, ERROR STATUS
1437 7301      CLA CLL IAC      /ENABLE CLEAR CONTROL
1440 4445      CLRALL      /CLEAR CONTROL
1441 2200      ISZ      RESTOR      /UPDATE FOR GOOD RECALIBRATE
1442 5600      JMP I      RESTOR      /RETURN
1443 4437      RESTER, ERROR      /ERROR, STATUS
1444 3603      J603      /TEXT POINTER

1445 4452      /      CRLF
1446 4447      PRNTER      /PRINT "TRY RECALIBRATE"
1447 1756      ERMES2
1450 4434      RECEIV
1451 5254      JMP      *-5
1452 5245      JMP      *-5
1453 5201      JMP      RESTOR +1
1454 7301      CLA CLL IAC      /WAIT FOR INPUT
1455 1056      TAD      AMOUNT      /TRY NEXT EXISTING DISK
1456 7450      SNA
1457 5535      JMP I      XEND      /GET AMOUNT ON SYSTEM
1460 3056      DCA      AMOUNT      /WAS THERE ONLY 1 LEFT
1461 3516      DCA I      TCNTRS      /LAST DISK
1462 5600      JMP I      RESTOR      /MORE TO GO BUT CLEAR THIS ONE
1463 0000      /      /ROUTINE TO CHANGE DEVICE CODES
1464 4777      CHANG, 0      JMS      XC0SW      /GET SWITCH REGISTER BITS.
1465 7010      RAR

```

```

1466 7620      SNL CLA      /CHANGE DEVICE CODES?
1467 5663      JMP I      CHANG      /NO.
1470 4777      JMS      XC0SW      /GET SWITCHES,
1471 0313      AND      A0770
1472 3314      DCA      CSAVE1      /SAVE DESIRED
1473 1316      TAD      CCNTR1
1474 3315      DCA      CSAVE2
1475 1317      TAD      CHNPOT
1476 3200      DCA      RESTOR
1477 1600      CHANG, TAD I      RESTOR      /GET ADDRESS POINTER
1500 3311      DCA      KWAIT
1501 1711      TAD I      KWAIT      /GET OLD CODE
1502 0312      AND      A7007      /MASK
1503 1314      TAD      CSAVE1      /ADD IN DESIRED
1504 3711      DCA I      KWAIT      /STORE DESIRED DEVICE CODE
1505 2200      ISZ      RESTOR      /UPDATE POINTER
1506 2315      ISZ      CSAVE2      /UPDATE CHANGE COUNTER
1507 5277      JMP      CHANGR
1510 5663      JMP I      CHANG      /EXIT TO PROGRAM.

1511 0000      KWAIT, 0
1512 7007      A7007, 7007
1513 0770      A0770, 0770
1514 0000      CSAVE1, 0
1515 0000      CSAVE2, 0
1516 7771      CCNTR1, 7771
1517 1520      CHNPOT, CHNPOT +1
1520 0734      IOT0
1521 0741      IOT1
1522 0746      IOT2
1523 0714      IOT3
1524 0705      IOT4
1525 0672      IOT5
1526 0727      IOT6
1527 2000      K2000, 2000

/      /THIS ROUTINE WILL GENERATE TIMING IF NEEDED BY THE APT SYSTEM
/      /
1530 0000      KTICK, 0
1531 7300      CLL CLA
1532 1022      TAD      22
1533 0070      AND      K4000      /GET HARDWARE CONFIGURATION
1534 7650      SNA CLA
1535 5351      JMP      EXTICK      /ON APT?
1536 1730      TAD I      KTICK      /NO
1537 3353      DCA      COUNT      /GET TIMING VALUE
1540 2132      ISZ      CLKCNT      /ESTABLISH TIME
1541 5351      JMP      EXTICK      /RETURN
1542 1353      TAD      COUNT      /GET VALUE OF COUNTER
1543 3132      DCA      CLKCNT      /STORE IT
1544 2354      ISZ      CNT
1545 5351      JMP      EXTICK      /TIMING NEED BE DONE?
1546 4425      TIME
1547 1355      TAD      KCNT
1550 3354      DCA      CNT      /TIMING VALUE
1550 3354      /INIT SECOND COUNTER

```

```

1551 2330 EXTICK, ISZ KTICK
1552 5730 JMP I KTICK /MOVE BEYOND TIMING VALUE

1553 0000 COUNT, 0
1554 7776 CNT, -2
1555 7776 KCNT, -2
1556 0100 K0100, 0100
/
/
/ROUTINE TO NOTIFY APT OF USE IF REQUIRED
/
1557 0000 KTIME, 0
1558 0002 IOF /DISABLE INTERRUPTS
1559 0214 RUF /GET PRESENT DATA FIELD
1560 1075 TAD KCUF
1561 3364 DCA ,+1 /ESTABLISHES CURRENT DATA FIELD
1562 7402 HLT
1563 6272 CIF 70 /FIELD 7, LOCATION OF UV FROM
1564 4771 JMS I K6500
1565 7300 CLL CLA
1566 5757 JMP I KTIME
/
1571 6500 K6500, 6500
/
1577 3062 PAGE
1578 1600 /
/
/THIS ROUTINE WILL NOTIFY APT OF AN ERROR AND SEND PC TO
/APT SYSTEM. ALL ERRORS WILL RESULT IN PROGRAM HLT AND A TIME OUT ON
/APT. APT WILL TAKE OVER FROM THERE.
/
1600 0000 AENR0, 0
1601 0002 IOF /DISABLE INTERRUPTS
1602 7200 CLA
1603 1022 TAD /CHECK FOR APT SYSTEM
1604 7700 SMA CLA
1605 5600 JMP I AENR0 /RETURN NOT ON APT
1606 1621 TAD I KENR0 /GET PC
1607 3222 DCA SAVPC
1608 6214 RUF /GET CURRENT DATA FIELD
1609 1075 TAD KCUF
1610 3214 DCA ,+2
1611 1222 TAD SAVPC
1612 7402 HLT /REPLACES WILL CURRENT DATA FIELD
1613 6272 CIF 70 /CHANGE IF FOR APT RETURN TO FIELD 7
1614 5620 JMP I K6520 /NOTIFIES APT OF ERROR
1615 7402 HLT
/
1620 6520 K6520, 6520
1621 0436 KENR0, ERNO
1622 0000 SAVPC, 0
/
/

```

```

/ROUTINE TO MOVE DISK POINTERS
/
1623 0000 MOVE, 0
1624 1237 TAD ADPT1
1625 5010 DCA AUTO10

1626 1240 TAD ADPT2
1627 5011 DCA AUTO11
1628 1077 TAD M10
1629 3241 DCA MCNTR1
1630 1410 TAD I AUTO10 /FROM HERE
1631 3411 DCA I AUTO11 /TO THERE
1632 2241 ISZ MCNTR1 /4 POINTERS
1633 5232 JMP ,+5
1634 5623 JMP I MOVE
/
1637 0137 ADPT1, DSK0A =1
1640 0150 ADPT2, DSK0B =1
1641 0000 MCNTR1, 0
/
/
1642 2003 TEXPC, TEXT "PC:"
1643 7200
1644 0704 TEXGO, TEXT "GO:"
1645 7200
1646 0530 TEXEX, TEXT "EX:"
1647 7200
1648 0315 TEXCH, TEXT "CH:"
1649 7200
1650 2324 TEXST, TEXT "ST:"
1651 7200
1652 0401 TEXOA, TEXT "OA:"
1653 7200
1654 0301 TEXCA, TEXT "CA:"
1655 7200
1656 0104 TEXAD, TEXT "AD:"
1657 7200
1658 0424 TEXDT, TEXT "DT:"
1659 7200
/
1664 2205 ERTX1, TEXT "READ STATUS ERROR"
1665 0104
1666 4023
1667 2401
1668 2425
1669 2340
1670 0522
1671 2217
1672 2200
1673 0411 ERTX2, TEXT "DISK DATA ERROR"
1674 2313
1675 4004
1676 0124
1677 0140
1678 0522

```

```
1703 2217
1704 2200
1705 2722      ERTX3, TEXT      "WRITE STATUS ERROR"
1706 1124
1707 0540
1710 2324
1711 0124
1712 2523
1713 4005
1714 2222
1715 1722
1716 0000
1717 2205      ERTX4, TEXT      "RECALIBRATE STATUS ERROR"
1720 0301
1721 1411
1722 0222
1723 0124
1724 0540
1725 2324
1726 0124
1727 2523
1730 4005
1731 2222
1732 1722
1733 0000

1734 2422      /
1735 3140      ERMES1, TEXT "TRY TO FORMAT SAME CYLINDER AGAIN?"
1736 2417
1737 4006
1740 1722
1741 1501
1742 2440
1743 2301
1744 1505
1745 4003
1746 3114
1747 1116
1750 0405
1751 2240
1752 0107
1753 0111
1754 1677
1755 0000
1756 2422      ERMES2, TEXT "TRY TO RECALIBRATE SAME DISK AGAIN?"
1757 3140
1760 2417
1761 4022
1762 0503
1763 0114
1764 1102
1765 2201
1766 2405
1767 4023
1770 0115
```

```
1771 0540
1772 0411
1773 2313
1774 4001
1775 0701
1776 1116
1777 7700
2000 2422      ERMES3, TEXT "TRY TO CHECK SAME CYLINDER AGAIN?"
2001 3140
2002 2417
2003 4003
2004 1005
2005 0313
2006 4023
2007 0115
2010 0540
2011 0331
2012 1411
2013 1604
2014 0522
2015 4001
2016 0701
2017 1116
2020 7700

2021 2213      /
2022 7005      TEXEND, TEXT      "RK8E/RK8L DISK FORMATTER PASS COMPLETE"
2023 5722
2024 1370
2025 1440
2026 0411
2027 2313
2030 4006
2031 1722
2032 1501
2033 2424
2034 0522
2035 4020
2036 0123
2037 2340
2040 0317
2041 1520
2042 1405
2043 2405
2044 0000
2045 2213      MES1, TEXT "RK8E/RK8L DISK FORMATTER PROGRAM"
2046 7005
2047 5722
2050 1370
2051 1440
2052 0411
2053 2313
2054 4006
2055 1722
2056 1501
```

```

2057 2424
2060 0522
2061 4020
2062 2217
2063 0722
2064 0115
2065 0000
2066 0617 MES2, TEXT "FOR ALL QUESTIONS, ANSWER Y FOR YES OR N FOR NO."
2067 2240
2070 0114
2071 1440
2072 2125
2073 0523
2074 2411
2075 1716
2076 2354
2077 4001
2100 1623
2101 2705
2102 2240
2103 3140
2104 0617
2105 2240
2106 3105
2107 2340
2110 1722
2111 4016
2112 4006
2113 1722
2114 4016
2115 1756
2116 0000
2117 0617 MES3, TEXT "FORMAT DISK "
2120 2215
2121 0124
2122 4004
2123 1123
2124 1340
2125 0000
2126 0122 MES4, TEXT "ARE YOU SURE?"
2127 0540
2130 3117
2131 2540
2132 2325
2133 2205
2134 7700
2135 0617 MES5, TEXT "FORMAT SAME DISK(S) AGAIN?"
2136 2215
2137 0124
2140 4023
2141 0115
2142 0540
2143 0411
2144 2313
2145 5023

```

```

2146 5140
2147 0107
2150 0111
2151 1677
2152 0000
      /
2200 PAGE
      /
2200 WRKBUF=.
      /
2200 M1TRK=.
2201 LDTRK=, +1
      /
2577 ENDBUF=, +377
      /

/CONSOL SRC =V2-RB= CONSOLE PACKAGE

/LAS= CALL C0CASH OR JMS XC88W
/THIS WILL READ THE SWITCH REGISTER FROM THE PLACE SPECIFIED
/BY LOCATION 20 BIT 0.

/THE PROGRAM SHOULD CHECK FOR A CONTROL CHARACTER FROM THE TERMINAL
/EVERY FIVE(5) SECONDS OR SOONER.

/LOCATIONS THAT NEED TO BE SET UP FOR USING THE CONSOLE PACKAGE.

/CNTVAL IN XC8PASS THIS LOCATION DETERMINES THE NUMBER OF
/PROGRAM COMPLETIONS THAT ARE NEEDED BEFORE THE PASS MESSAGE IS TYPED
/THE VALUE SHOULD PUT THE PASS MESSAGE OUT IN THE RANGE OF 1 TO 5 MINUTES.
/THIS SHOULD BE A POSITIVE NUMBER.

/C08TR1 THIS IS FOUND IN CNTNL ROUTINE CONTROL R PART
/IT IS THE RETURN WHEN CONTROL R IS ENTERED (RESTART PROGRAM)
/THE RETURN JUMPS TO X008W WHICH CONTAINS C08TR1 SO PUT THE LABEL C08TR1
/WHERE YOU WANT TO RESTART THE PROGRAM.

/SETUP1 IN XC8ERR THIS IS THE MASK BIT FOR HALT ON ERROR
/PLACE THE CORRECT BIT IN THIS LOCATION FOR HALTING ON ERRORS.

/SETUP2 IN XC8PASS THIS IS THE MASK FOR HALT A END OF PASS.

/THE CALL TABLE IS A CONDITIONAL ASSEMBLY.
/TO ASSEMBLE THE CALL REMOVE THE / BEFORE CONSOL=0.
/IN COMBINING THE CONSOL PACKAGE TO A DIAGNOSTIC.
/THE CALL TABLE IS TO BE AT THE BEGINNING OF A PROGRAM.

/CONSOL=0
6661 PSKF= 6661
6662 PCLF= 6662

```

```

6663      PSKE= 6663
6664      PSTB= 6664
6665      PSIE= 6665
6004      GTF= 6004
7701      ACL= 7701
6007      CAF= 6007
7421      MQL= 7421
7501      MQA= 7501
/
3000      *3000
/
/*****
/CBPASS
/THIS IS CALLED AT THE END OF EACH PROGRAM COMPLETION
/THE VALUE OF** CNTVAL** WILL BE DETERMINED BY THE TIME IT TAKES
/THE PROGRAM TO COMPLETE THIS MANY CBPASS TO BE IN THE 1 TO 4 MINUTE
/RANGE
/      CBPASS=JMS      XCBPAS
/EX. OF CALL      CBPASS
/      /      HLT      /HALT IF NON CONSOL PACKAGE
/      JMP      START1 /CONTINUE RUNNING THIS PROGRAM

/RETURN TO LOCATION CALL PLUS ONE WITH THE AC=0 IF NON CONSOL PACKAGE AND HLT
/IF CONTINUE TO RUN THEN RETURN TO CALL PLUS2 AC=0
/THE LOCATION SETUP2 IS THE MASK BIT FOR THE HALT AT END OF PASS
/CHECK THAT IT IS CORRECT FOR THE CURRENT PROGRAM

/CALLS USED BY XCBPAS ARE CHKCLA-XC8CLF-XC8OCTA-XC8SW-XC8PNT-XC8ING-

```

```

3000 0000 XCBPAS, 0
3001 7200 CLA
3002 4777* JMS CHKCLA /IS WORD 22 BIT 3 ACTIVE CONSOLE?
3003 5212 JMP DOPACK /IS CLASSIC
3004 4776* JMS C8GET /GET THE REGISTERS.
3005 4262 JMS XC8SW /DEACTIVE CONSOL CHECK SR SETTING
3006 0375 AND 1400 /FOR HALT ON END OF CBPASS
3007 7640 SZA CLA /1= HALT 0 CONTINUE
3010 5600 JMP I XCBPAS /GO TO HALT
3011 5230 JMP C8BY1 /CONTINUE ON RUNNING PROGRAM
3012 4232 DOPACK, JMS CKCOUT /CLASS CHECK CBPASS COUNT
3013 5230 JMP C8BY1 /CBPASS COUNT NOT DONE REDO PROGRAM
3014 2250 ISZ PASCNT /CBPASS COUNT DONE SET CBPASS COUNT
3015 4774* JMS XC8CLF
3016 4303 JMS XC8PNT /CBPNT BUFFER
3017 3053 MESPAS /
3020 1250 TAD PASCNT /GET NUMBER
3021 4773* JMS XC8OCTA /CONVERT IT TO ASCII
3022 4774* JMS XC8CLF /DO A CARRIAGE RETURN
3023 4776* JMS C8GET /GET THE REGISTERS.
3024 4262 JMS XC8SW /CHECK A HALT AT END OF CBPASS
3025 0375 SETUP2, AND 1400 /MASK BIT
3026 7640 SZA CLA /HALT =1 NO SKIP CONTINUE =0
3027 4772* JMS XC8ING /STOP PROGRAM EXECUTION-LOOK FOR INPUT

```

```

3030 2200 C8BY1, ISZ XCBPAS /BUMP RETURN
3031 5600 JMP I XCBPAS
3032 0000 CKCOUT, 0
3033 1251 TAD DOSET /CHECK IF SET UP NEEDED
3034 7640 SZA CLA /0=SET UP CBPASS COUNT VALUE
/1=CBPASS COUNT VALUE OK
3035 5242 JMP NOSET /CBPASS COUNT VALUE OK
3036 1252 TAD CNTVAL /GET COUNT VALUE FOR THIS PROG
3037 7040 CMA /SET TO NEGATIVE
3040 3247 UCA DOCNT /STONE IN HERE
3041 2251 ISZ DOSET /INDICATE VALUE SET UP
3042 2247 NOSET, ISZ DOCNT /COUNT THE NUMBER OF PASSES
3043 5230 JMP C8BY1 /EXIT FOR ANOTHER PASS
3044 3251 DCA DOSET /SET TO CBPNT CBPASS
3045 2232 ISZ CKCOUT /BUMP RETURN FOR
3046 5632 JMP I CKCOUT /CBPASS C8TYPE OUT
3047 0000 DOCNT, 0 /
3050 0000 PASCNT, 0
3051 0000 DOSET, 0
3052 0000 CNTVAL, 0
3053 0410 MESPAS, TEXT "DNRKD PASS "
3054 2213
3055 0404
3056 4040
3057 2001
3060 2323
3061 4000

```

```

/*****

```

```

/C8CKSW

```

```

/THIS ROUTINE CAN BE USED INPLACE OF A READ THE SWITCHES LAS.
/ROUTINE THAT WILL CHECK WHERE TO READ THE
/C8 SWITCHES FROM IE. FROM PANEL OR PSEUDO SWITCH REGISTER
/THE SELECTION IS DETERMINED BY THE STATE OF BIT 0 IN LOCATION 21.

```

```

/C8CKSW= JMS XC8SW
/EX. JMS XC8SW /READ THE C8SWIT REGISTER
/RETURN WITH THE CONTENTS OF SWITCH REGISTER

```

```

/RETURN TO NEXT LOCATION FOLLOWING CALL WITH THE AC= TO VALUE OF C8SWIT SETTING

```

```

/CALLS USED ARE-XC8CKPA-

```

```

3062 0000 XC8SW, 0
3063 4771* JMS XC8CKPA /GO CHECK THE IF ANY CONTRL
3064 7000 NOP
3065 1021 TAD 21 /GET WD FOR INDICATOR
3066 7710 SPA CLA /CHECK IF FROM PANEL 4000
3067 7614 /DO LAS AND SKIP GET FROM PANEL WITH LAS

```

```

3070 1020      TAD      20      /PSEUDU SWITCH
3071 5662      JMP I   XC8SW    /EXIT WITH STATUS BIT IN AC.

/*****
/C8TTYI
/THIS ROUTINE WILL LOOK FOR A INPUT FROM THE TERMINAL
/AND REMOVE ANY PARITY BITS, THEN MAKE IT 8 BIT ASCII.
/      C8TTYI= JMS XC8TTY
/EX.      JMS      XC8TTYI      /READ CHAR FROM THE CONSOL DEVICE
/      /                      /RETURN TO CALL PLUS ONE AC CONTAINS THE CHAR

/CALLS USED =NONE= BUT C8CHAR IS OFF PAGE AND IN ROUTINE CALLED XC8ECHO
/
/
3072 0000      XC8TTY, 0
3073 6031      KSF                      /LOOK FOR KEYBOARD FLAG
3074 5273      JMP      .-1
3075 6036      KRB                      /GET CHAR
3076 0370      AND      (177          /MASK FOR 7 BITS
3077 1367      TAD      (200         /ADD THE EIGHTH BIT
3100 3766      OCA      C8CHAR       /STORE IT
3101 1766      TAD      C8CHAR
3102 5672      JMP I   XC8TTY        /EXIT

```

```

/*****
/C8PRNT
/THIS ROUTINE WILL TYPE THE CONTENTS OF THE C8 PRINT BUFFER, THE LOCATION
/OF THE BUFFER WILL BE IN THE ADDR5 FOLLOWING THE CALL. PRINTING OF THE BUFFER
/WILL STOP WHEN A 00 CHAR IS DETECTED. CHARACTERS ARE PACKED 2 PER WORD.
/      C8PRNT= JMS XC8PNT

/EX.      JMS      XC8PNT      /C8PRNT THE CONTENTS OF THE FOLLOWING BUFFER
/      MESS77              /LOCATION OF C8PRNT BUFFER

/C8PRNT WILL USE THE LOCATION FOLLOWING THE CALL AS THE POINTER FOR THE
/C8PRNT ROUTINE, RETURN TO CALL PLUS TWO WITH AC= 0

/CALLS USED ARE=XC8TYPE=XC8PNT

```

```

3103 0000      XC8PNT, 0
3104 7300      CLA CLL
3105 1703      TAD I   XC8PNT      /GET C8PNT BUFFERS STARTING LOCATION
3106 3336      OCA      PTSTOR     /STORE IN PTSTOR

```

```

3107 2303      C8D01, ISZ      XC8PNT      /BUMP RETURN
3110 1736      TAD I   PTSTOR     /GET DATA WORD
3111 0365      AND      (7700     /MASK FOR LEFT BYTE
3112 7450      SNA                      /CHECK IF 00 TERMINATE
3113 5703      JMP I   XC8PNT      /EXIT
3114 7500      SMA                      /IS AC MINUS
3115 7020      CML                      /MAKE CHAR A 300 AFTER ROTATE
3116 7001      IAC                      /MAKE CHAR A 200 AFTER ROTATE
3117 7012      RTR
3120 7012      RTR
3121 7012      RTR
3122 4764      JMS      XC8TYPE     /PUT CHAR IN BITS 4-11 MAKE IT 8 BIT ASCII
3123 1736      TAD I   PTSTOR     /C8PNT IT ON CONSOLE
3124 0363      AND      (0077     /GET DATA WORD
3125 7450      SNA                      /MASK FOR RIGHT BYTE
3126 5703      JMP I   XC8PNT      /CHECK IF 00 TERMINATOR
3127 1362      TAD      (3740     //EXIT
3130 7500      SMA                      /ADD FUDGE FACTOR TO DETERMINE IF 200
3131 1361      TAD      (100       /OR 300 IS TO BE ADD TO CHAR
3132 1360      TAD      (240       /ADD 100
3133 4764      JMS      XC8TYPE     /ADD 200
3134 2336      ISZ      PTSTOR     /C8TYPE ONLY BITS 4-11
3135 5310      JMP      C8D01      /BUMP POINTER FOR NEXT WORD
3136 0000      PTSTOR, 0          /DO AGAIN
/STOK FOR C8PRNT BUFFER

/*****
/C8PAUS
/THIS ROUTINE WILL CHECK IF THE CONSOL PACKAGE IS ACTIVE, IF ACTIVE
/IT WILL RETURN TO CALL PLUS ONE AC= 0. AND DO THAT INSTRUCTION.
/IF THE CONSOL PACKAGE IS NOT ACTIVE THE CALL WILL BE REPLACED
/WITH A 7402 HALT AND THEN RETURN TO THE HALT.
/      C8PAUS= JMS XC8PAU
/
/EX.      JMS      XC8PAUS      /CHECK IF ON ACTIVE CONSOL IF NOT HALT HERE
/      ANYTHING              /RETURN HERE IF ON ACTIVE CONSOL
/
/
/CALLS USED ARE =CHKCLA=

3137 0000      XC8PAU, 0
3140 7300      CLA CLL
3141 4777      JMS      CHKCLA     /CHECK LOC 22 BIT 3 CONSOLE BIT
3142 5350      JMP      C8D03      /GO DO CONSOL PART RETURN CALL +1
3143 7040      CMA                      /DEACTIVE CONSOLE PACKAGE PUT HLT IN CALL
3144 1337      TAD      XC8PAU     /GET CORRECT RETURN ADDR5
3145 3337      OCA      XC8PAU     /SET UP RETURN
3146 1357      TAD      (7402     /GET CODE FOR HLT
3147 3737      DCA I   XC8PAU     /PUT HLT IN CALL LOCATION
3150 5737      C8D03, JMP I   XC8PAU /GO TO HALT OR RETURN TO NEXT LOCATION

```

3157 7402
3160 0240
3161 0100
3162 3740
3163 0077
3164 3677
3165 7700
3166 3675
3167 0200
3170 0177
3171 3641
3172 3435
3173 3600
3174 3623
3175 0400
3176 3424
3177 4000
3200

PAGE

/*****

/C0CNTN

/THIS ROUTINE WILL CHECK FOR THE PRESENCE OF CONTROL CHARACTERS

/IT WILL CHECK FOR THE FOLLOWING CHAR C-R-Q-L-S

/ C0CNTN= JMS XC0CNT

/EX.

JMS XC0CNTN

/CHECK FOR CONTROL CHARACTER

/

JMP ANYTHING

/LUC FOLLOWING CALL IS FOR CONTINUING THE PROGRAM

/

JMP ANYTHING

/LUC. IS FOR RETURN IF INMODE SET AND NOT CNTRL CHAR

/RETURN IS TO CALL PLUS ONE IF CONTINUE

/RETURN IS TO CALL PLUS TWO IF INMODE SET AND NOT CONTROL CHAR

/RETURN IS TO CALL PLUS TWO IF INMODE IS NOT SET AND NO

/CNTRL CHAR ..THIS WILL PRINT THE CHARACTER AND A ?

/CLEAR THE AC AND RETURN CALL+2.

/CALLS USED ARE=CHKCLA=XC0TYPE=XC0CNLF=C0GET=UPAKOW=XC0TYI=XC0PSW=

/

/

XC0CNT, 0

3200 0000
3201 3777
3202 4776
3203 5206
3204 1777
3205 5600
3206 6004
3207 3775
3210 7501
3211 3774
3212 3255
3213 1257

DCA ACSAVE

/SAVE THE AC

JMS CHKCLA

/CHECK LUC.22 BITS FOR CONSOLE BIT

JMP

/ON ACTIVE CONSOLE

TAD ACSAVE

/DEACTIVE CONSOLEGET AC FOR RETURN

JMP I XC0CNT

/EXIT NOT ON ACTIVE CONSOLE

GTF

DCA FLSAVE

DCA MQA

DCA MQSAVE

/SAVE THE MQ

DCA INDEXA

/SET DISPLACEMENT INTO TABLE B

TAD XTABLA

/GET ADDRS OF TABLE A

3214 3256
3215 1656
3216 7450
3217 5226
3220 1773
3221 7650
3222 5243
3223 2255
3224 2256
3225 5215
3226 1772
3227 7640
3230 5240
3231 1773
3232 4771
3233 1370
3234 4771
3235 4767
3236 2200
3237 5600
3240 2200
3241 1773
3242 5600
3243 1773
3244 1366
3245 3773
3246 1260
3247 1255
3250 3254
3251 1654
3252 3254
3253 5654
3254 0000
3255 0000
3256 0000
3257 3261
3260 3271
3261 7575
3262 7564
3263 7557
3264 7556
3265 7555
3266 7573
3267 7574
3270 0000

3271 3347
3272 3336
3273 3300
3274 3311
3275 3320
3276 3344
3277 3400

REDDA,

DCA GETDAT

/CONTAINS POINTER TO CONTROL CHAR

TAD I GETDAT

/GET CNTRL CHAR FROM TABLE

SNA

/CHECK FOR A 0 END OF TABLE

JMP DONEA

/END OF TABLE NO CONTROL CHAR

TAD C0CHAR

/COMPARE CHAR TO CONTROL CHAR

SNA CLA

/B IF MATCH

JMP GOITA

/MATCH

ISZ INDEXA

/NO MATCH NOT END OF TABLE REDD

ISZ GETDAT

/BUMP INDEX FOR EXIT WHEN CONTROL FOUND

JMP REDDA

/BUMP GETDAT FOR COMPARE OF NEXT CNTRL CHAR.

TAD INMODE

/CHECK IF PROGRAM EXPECTS CHAR

SZA CLA

/1=CHAR EXPECTED 0= NO CHAR EXPECTED

JMP EXITA

/CHAR EXPECTED

TAD C0CHAR

/GET CHAR = NOT CONTROL + NOT EXPECTED

JMS XC0TYPE

/C0PMNT CHAR

TAD (277

/GET CODE FOR "Z"

JMS XC0TYPE

JMS XC0CNLF

/BUMP RETURN

ISZ XC0CNT

/EXIT CALL+2

JMP I XC0CNT

/BUMP RETURN FOR MAIN PROGRAM CHECK OF CHAR

ISZ XC0CNT

/PUT CHAR IN AC.

TAD C0CHAR

/EXIT

JMP I XC0CNT

/GET THE CONTENTS OF CHAR

TAD C0CHAR

/ADD 100 TO FORM A GOOD ASCII CHARACTER

TAD (100

/RESTORE CORRECT CHAR

DCA C0CHAR

/GET START OF TABLE B

TAD XTABLB

/GET NOW FAR INTO TABLE

TAD INDEXA

/STORE IT

DCA G0TOA

/GET THE ROUTINE STARTTING ADDRESS

TAD I G0TOA

/STORE IT IN HERE

DCA G0TOA

/GOTO CONTROL CHAR ROUTINE

JMP I G0TOA

/ADD UP CNTRL ROUTINE TO EXECUTE

G0TOA, 0000

/DISPLACEMENT INTO CNTRL TABLE

INDEXA, 0000

/LOCATION OF ADDRS OF CONTROL CHAR.

GETDAT, 0000

/ADDNS OF TABLEA

XTABLA, TABLA

/ADDNS OF TABLEB

XTABLB, TABLB

/CNTRL C BACK TO MONITOR 203

TABLA, 7575

/CNTRL L SWITCH ERROR PRINTING DEVICE 214

7564

/CNTRL Q START DISPLAYING CHAR. AGAIN 221

7557

/CNTRL R BACK TO BEGINNING OF PROGRAM 222

7556

/CNTRL S STOP SENDING CHAR TO DISPLAY WAIT FOR CNTRL Q 223

7555

/CNTRL E CONTINUE WITH PROGRAM 205

7573

/CNTRL D CHANGE SWITCH REGISTER ON FLY

7574

0000

TABLB, CNTRLC

CNTRLQ

CNTRLR

CNTRLB

CNTRLE

CNTRLD

/

/CNTRL Q

```

/START SENDING CHAR. TO THE DISPLAY
/THIS WILL RETURN CONTROL TO CALL THAT WAS SET BY
/THE CALL FOR CONTROL S.
/
3300 3772* CNTRLQ, DCA INMODE /SET SUFT FLAG FOR UNEXPECTED CHAN
3301 1334 TAD C0SETS /CHECK IF CONTROL S TYPED IN
3302 7640 SZA CLA
3303 5306 JMP BYMETR /CONTROL S TYPED IN
3304 4765* JMS C0GET /NO CONTROL S TYPED PREVIOUSLY
3305 5600 JMP I XC0CNTR /LEAVE VIA CNTR ENTRY ADDRESS
3306 3334 BYMETR, DCA C0SETS /CLEAR THE SUFT FLAG
3307 4765* JMS C0GET /RESTORE REGISTERS
3310 5735 JMP I C0METR /EXIT TO ADDRESS SET BY CONTROL S
/
/
/CONTROL R
/GO TO THE QUESTION C0SWIT
3311 3764* CNTRLR, DCA TTYLPT /CLEAR THE TYPE FLAG SET TO TTY
3312 3334 DCA C0SETS /CLEAR SUFT FLAG FOR CNTRL S
3313 3772* DCA INMODE
3314 4763* JMS UPAROW /PRINT THE " AND C0CHAN
3315 3762* C0BY4, DCA C0SWT /CLEAR FLAG FOR CNTRL D OR N
3316 5717 JMP I XD0SW /GO TO ADDRS OF C0SWIT
3317 0200 XD0SW, BGN /D0SW IS LABEL FOR C0SWIT QUESTION
/
/
/CONTROL S
/STOP SENDING CHAR. TO DISPLAY UNTIL A "U IS RECEIVED
/
/
3320 1334 CNTRL3, TAD C0SETS /IF1 DU NOT STORE IN C0RETR
3321 7640 SZA CLA
3322 5326 JMP C0U07 /DONT SET UP C0METR
3323 7001 IAC /MAKE RETURN CALL PLUS 2
3324 1200 TAD XC0CNT /GET RETURN FOR THIS CALL
3325 3335 DCA C0METR /STORE IT HERE FOR USE BE CNTRL U
3326 2334 C0U07, ISZ C0SETS /SET FLAG TO SAVE CALL
3327 4761* JMS XC0TTYI /LOOK FOR THE INPUT
3328 4765* JMS C0GET /GET REGISTERS
3331 4200 JMS XC0CNTR /CHECK FOR THE CONTROL CHAR
3332 7200 CLA
3333 5320 JMP CNTRL3 /IF NOT A CNTRL Q R C REASK
3334 0000 C0SETS, 0
3335 0000 C0METR, 0
/
/SWITCH OUTPUT FROM ONE OUTPUT DEVICE TO ANOTHER - THE TWO OUTPUTS ARE THE
/CONSOLE AND THE PRINTER WITH DEVICE CODE 06.
/
/
3336 1764* CNTRL4, TAD TTYLPT /GET PRESENT C0SWIT INVICATUR
3337 7040 CMA /COMPLEMENT IT
3340 3764* DCA TTYLPT /STORE NEW C0SWIT
3341 4763* JMS UPAROW /C0PHNT " AND CHAR ON NEW DEVICE
3342 4765* JMS C0GET /RESTORE THE REGISTERS
3343 5600 JMP I XC0CNT /EXIT

```

```

/
/CONTROL E
/CONTINUE RUNNING FROM A INQUIRE OR ERROR
/
/
3344 4763* CNTRL5, JMS UPAROW /PRINT THE CONTROL CHAR
3345 4765* JMS C0GET /GET THE REGISTERS
3346 5600 JMP I XC0CNT /RETURN TO CALL PLUS ONE
/
/
/CONTROL C
/RETURN TO MONITOR CONTROL C
3347 3764* CNTRL6, DCA TTYLPT /CLEAR THE LPT FLAG TO PRINT ON DISPLAY
3350 4763* JMS UPAROW /C0PHNT " AND LETTER IN CHAR
3351 6203 CDF CIP /GO TO B FLD
3352 0007 CAF /CLEAR THE WORLD
3353 5760 JMP I L7600 /GO TO DIAGNOSTIC MONITOR
/*****
/
/
3360 7600
3361 3072
3362 3545
3363 3415
3364 3721
3365 3424
3366 0100
3367 3023
3370 0277
3371 3077
3372 3076
3373 3075
3374 4123
3375 4124
3376 4000
3377 4122
3400 PAGE
/
/CONTROL D
/CHANGE THE SWITCH REGISTER ANYTIME CNTRL D AND RETURN TO
/THE PROGRAM RUNNING.
3400 4215 CNTRL7, JMS UPAROW
3401 1213 TAD C0SET0 /CHECK IF THE RETURN ADDRS IS SAFE
3402 7640 SZA CLA
3403 5207 JMP C0U011 /DO NOT CHANGE THE RETURN ADDRS
3404 1777* TAD XC0CNT /GET THE RETURN ADDRS AND SAVE IT
3405 3214 DCA C0RETD /SAVE THE RETURN HERE
3406 2213 ISZ C0RETD /INDICATE RETURN SAVED DONT DESTROY
3407 4256 C0U011, JMS XC0PSW /GO CHANGE THE SWITCH REGISTER
3410 3213 DCA C0RETD /CLEAR THE FLAG
3411 4224 JMS C0GET /RESTORE THE AC HQ LINK ETC

```

```

3412 5614      /      JMP I  C8NETD      /RETURN TO THE PROGRAM
3413 0000      C8NETD, 0
3414 0000      C8NETD, 0

/THIS WILL TYPE A UP ARROW AND THE CHAR IN C8CHAR.

3415 0000      UPAROW, 0      /C8PNT THE "-" AND THE CHAR C8TYPED IN
3416 1376      TAD      C336      /CODE FOR "-"
3417 4775      JMS      XC8TYPE
3420 1774      TAD      C8CHAR      /C8TYPE THE CHAR
3421 4775      JMS      XC8TYPE
3422 4773      JMS      XC8CRLF
3423 5615      JMP I  UPAROW      /EXIT

```

```

/*****

3424 0000      C8GET, 0
3425 7200      CLA
3426 1772      TAD      MQSAVE
3427 7421      MQL      /RESTORE MQ
3430 1771      TAD      FLSAVE
3431 7004      HAL      /RESTORE THE LINK
3432 7200      CLA
3433 1770      TAD      ACSAVE
3434 5624      JMP I  C8GET      /RESTORE THE AC
                                /GET THE REGISTERS

```

/*****

```

/C8INQU
/C8INQU ROUTINE WILL PRINT A WAITING
/AND THE PROGRAM IS EXPECTING A CONTROL CHAR INPUT
/IF CONTINUE FROM CONTROL CHAR RETURN IS CALL PLUS ONE
/IF NO CONTROL CHAR ENTERED THEN WAITING IS REPRINTED
/AND PROGRAM WAITS FOR A CONTROL CHAR AGAIN.

```

```

/      C8INQU =      JMS XC8ING
/EX.      JMS      XC8ING      /C8 WILL PRINT A WAITING AND WAIT FOR INPUT
/      DO ANYTHING      /RETURN IS CALL PLUS ONE AC = 0 CONTINUE

```

/CALLS USED ARE -CHKCLA-XC8PNT-XC8TYI-C8GET-XC8CNTR-

```

3435 0000      XC8ING, 0
3436 7300      CLA CLL
3437 4767      JMS      CHKCLA      /CHECK LOC 22 BIT 3 CONSOLE BIT
3440 7410      SKP
3441 5635      JMP I  XC8ING      /ACTIVE CONSOLE PACKAGE
                                /NOT CONSOLE LEAVE

```

```

3442 4766      JMS      XC8PNT
3443 3451      WATHES
3444 4765      JMS      XC8TTYI      /INQUIRE WAITTING
3445 4224      JMS      C8GET      /GET CHARACTER
3446 4777      JMS      XC8CNTR      /CHECK IF CONTROL CHARACTER
3447 5635      JMP I  XC8ING      /EXIT AND CONTINUE
3450 5236      JMP      XC8ING+1      /REASK
3451 2701      WATHES, TEXT "WAITING "
3452 1124
3453 1116
3454 0740
3455 0000

```

/*****

/C8SWIT

```

/ROUTINE WILL CHECK IF CONSOL IS ACTIVE IF IT IS ACTIVE DISPLAY
/SW QUESTION, IN NOT ACTIVE IT WILL NOT PRINT THE SW QUESTION BUT
/RETURN TO CALL PLUS ONE AC=0,
/C8SWIT WILL SET UP THE PSEUDO SWITCH
/REGISTER WITH THE NEW DATA ENTERED

```

```

/      C8SWIT =      JMS XC8PSW
/EX.      JMS      XC8PSW      /SET UP PSEUDO C8SWIT REGISTER IF
                                /ON THE CONSOL PACKAGE. RETURN IS CALL PLUS ONE AC = 0

```

/CALLS USED ARE -CHKCLA-XC8PSW-XC8PNT-XC8OCTA-XC8TYPE-

```

3456 0000      XC8PSW, 0
3457 4767      JMS      CHKCLA      /CHECK LOC 22 BIT 3 CONSOLE BIT
3460 7410      SKP
3461 5656      JMP I  XC8PSW      /ACTIVE CONSOLE
                                /DEACTIVE CONSOLE PACKAGE
                                /RETURN WITHOUT ASKING PSEUDO SWITCH
3462 1345      TAD      C8SWST      /IS THE SW FLAG SET FOR SWITCH?
3463 7640      SZA CLA
3464 5764      JMP      C8NY4      /SKIP IF ONE ENTRY AT A TIME OK
3465 2345      ISZ      C8SWST      /SECOND ENTRY WITH OUT A EXIT GO TO SW QUESTION
3466 4766      JMS      XC8PNT      /FIRST ENTRY SET FLAG
3467 3547      MESA
3470 1020      TAD      20
3471 4763      JMS      XC8OCTA      /C8PNT SR=
3472 1362      TAD      140
3473 4775      JMS      XC8TYPE      /GET CONTENTS OF SW
3474 2761      ISZ      INMODE      /CONVERT IT TO ASCII
3475 4760      JMS      XC8ECHO      /GET SPACE
3476 4315      JMS      T8TCHA      /SET FLAG FOR CHAR EXECTED
3477 1774      TAD      C8CHAR      /LOOK FOR INPUT
3500 3020      DCA      20          /NOT CONTROL TEST IT IS LEGAL
                                /STORE NEW CHAR IN SW REG

3501 1357      TAD      (-3
3502 3346      DCA      T8PCNT      /GET A MINUS 3
                                /STORE IN TEMP COUNT

```

```

3503 4760*   GETCH1, JMS   XC8ECHU   /GET NEXT CHAN
3504 4315     JMS   TSTCHA   /CHECK IF CR + GOOD CHAR
3505 1020     TAD   20        /GET C8SWIT REGISTER
3506 7106     RTL   CLL      /ROTATE IT LEFT 3 PLACES
3507 7004     RAL
3510 1774*   TAD   C8CHAR   /GET CHAR + ADD IT TO PREVIOUS CONTENTS
3511 3020     DCA   20      /SAVE NEW CONTENTS
3512 2346     ISZ   TMPCNT   /BUMP COUNT
3513 5303     JMP   GETCH1   /JMP BACK + GET NEXT CHAR
3514 5342     JMP   ENUIT    /END 4 CHAR C8TYPED IN
3515 0000     TSTCHA, 0
3516 7041     CIA
3517 1356     TAD   (215    /CMPL CHAR IN AC
3520 7650     SNA   CLA     /TEST IF IT IS A CARRIAGE RETURN
3521 5342     JMP   ENUIT   /SKIP IN NOT CR.
3522 1774*   TAD   C8CHAR   /HAS CARRIAGE RETURN
3523 1355     TAD   (=260   /NOT CM, GET CHAR
3524 7710     SPA   CLA     /CHECK IF IT IS IN RANGE
3525 5336     JMP   ERR1    /IF NOT POSITIVE C8ERR CHAR SMALLER THEN 260
3526 1774*   TAD   C8CHAR   /C8ERR + CHAN TOO SMALL
3527 1354     TAD   (=270   /GET CHAN
3530 7700     SNA   CLA     /GET A -270 + CHECK IF IT IS LARGER THEN 7
3531 5336     JMP   ERR1    /SKIP IF LESS THEN 7
3532 1774*   TAD   C8CHAR   /C8ERR ON CHAR NOT IN RANGE
3533 0353     AND   (7      /GET CHAR
3534 3774*   DCA   C8CHAR   /MASK FOR RIGHT BYTE
3535 5715     JMP   I   TSTCHA /STORE IN CHAR
3536 1352     ERR1, TAD   (277 /GET CHAR IN AC
3537 4775*   JMS   XC8TYPE  /EXIT
3540 4773*   JMS   XC8CRLF  /C8PKNI
3541 5266     JMP   C8MDPS  /?
3542 4773*   ENUIT, JMS   XC8CRLF /EXIT + ASK AGAIN
3543 3345     DCA   C8SWST  /DO A CR LF
3544 5656     JMP   I   XC8PSW /CLEAR THE PSW ENTRY FLAG
3545 0000     C8SWST, 0     /EXIT ROUTINE

3546 0000     TMPCNT, 0
3547 2322     MESA, TEXT "SW "
3550 7540
3551 0000

```

```

3552 0277
3553 0007
3554 7510
3555 7520
3556 0215
3557 7775
3560 3663
3561 3676
3562 0040
3563 3600
3564 3315
3565 3072

```

```

3566 3103
3567 4000
3570 4122
3571 4124
3572 4123
3573 3623
3574 3675
3575 3677
3576 0336
3577 3200
3580 3600
3581 7106
3582 7006
3583 3221
3584 1377
3585 3222
3586 1221
3587 0376
3588 1375
3589 4277
3590 1221
3591 7006
3592 7004
3593 3221
3594 2222
3595 5206
3596 5600
3597 0000
3598 0000
3599 0000

```

PAGE

```

/C8OCTA

/OCTAL TO ASCII CONVERSION
/THIS ROUTINE WILL TAKE THE OCTAL NUMBER IN THE AC AND CONVERT IT TO ASCII
/THE RESULT WILL BE PRINTED ON THE CONSOL TERMINAL
/   C8OCTA* JMS XC8OCT
/
/EX.   JMS   XC8OCTA   /AC CONTAINS NUMBER TO BE CHANGE
/      RETURN IS TO CALL PLUS ONE AC=0
/
/CALLS USED ARE =XC8TYPE=

XC8OCT, 0
CLL RTL
RTL
DCA C8TMP1   /POSITION THE FIRST CHAR FOR PRINTING
TAD (=4     /SAVE CORRECT POSITIONED WORD HERE
DCA C8CKP   /STORE COUNTER IN HERE
TAD C8TMP1  /GET FIRST NUMBER
AND (0007   /MASK
TAD (260    /ADD THE PRINT CONSTANT
JMS XC8TYPE /TYPE THE NUMBER
TAD C8TMP1  /
RTL
RAL
DCA C8TMP1   /PUT NEXT NUMBER IN POSITION
ISZ C8CKP   /STORE IT
JMP C8OCT   /DONE YET WITH FOUR NUMBERS
JMP I   XC8OCT /NOT YET DO MORE
C8TMP1, 0   /DONE WITH FOUR
C8CKP, 0

```

/*****

```

/C8CRLF
/C8TYPE CR AND LF WITH FILLERS FOLLOWING EACH LF AND CR
/
/   C8CRLF* JMS XC8CRLF
/
/EX.   JMS   XC8CRLF   /C8PKNI A CR AND LF WITH FILL
/      /RETURN TO CALL PLUS ONE AC =0

```

/CALLS USED ARE =XC8TYPE=

```

3623 0000  XC8CRLF,0
3624 7300      CLA CLL
3625 1374      TAD      (215      /GET CODE FOR CR
3626 4277      JMS      XC8TYPE
3627 1237      TAD      FILLER
3630 7040      CHA
3631 3240      DCA      FILCNT      /STONE FILLER IN HERE
3632 1373      TAD      (212      /GET CODE FOR LF
3633 4277  C8UO2, JMS      XC8TYPE
3634 2240      ISZ      FILCNT      /CHECK ON FILLER CHAR
3635 5233      JMP      C8UO2      /TYPE A NON PRINTING CHAR
3636 5623      JMP I      XC8CRLF  /EXIT
3637 0004  FILLER, 0004      /FILLER SET FOR 4 CHAR
3640 0000  FILCNT, 0      /COUNTER FOR FILL

```

```

//*****
/C8CKPA
/THIS ROUTINE WILL CHECK IF A CHARACTER WAS ENTERED FROM THE
/TERMINAL. IF THE FLAG IS SET AND THE CONSOLE PACKAGE IS
/ACTIVE A CHECK IS MADE TO DETERMINE IF IT IS A CONTROL CHAR.
/IF IT WAS A CONTROL CHAR THEN ITS CONTROL FUNCTION IS PERFORMED.
/IF NOT A CONTROL CHARACTER OR A CONTROL E-D-L-D= IT WILL DO
/THE CONTROL FUNCTION AND RETURN TO CALL PLUS 2.
/IF A NON CONTROL CHARACTER WILL BE PRINTED AND A "?" IT WILL RETURN TO
/CALL PLUS 2.
/IF NO FLAG IS SET OR THE CONSUL IS NOT ACTIVE THE RETURN IS TO
/CALL PLUS 1.

```

/ C8CKPA= JMS XC8CKP

```

/EX.  JMS      XC8CKPA      /CALL TO CHECK IF CONTROL CHAR SET
/      ANYTHING(SKIP)      /RETURN IF NOT FLAG OR NOT CONSOLE ACTIVE
/      ANYTHING(JMP EXIT SKIP CHAIN) /RETURN IF NOT CONTROL OR CONTINUE CONTROL

```

/CALLS USED ARE =XC8TTYI=XC8CNTR=C8GET=

```

3641 0000  XC8CKP, 0
3642 3772  DCA      ACSAVE      /SAVE THE AC
3643 6004  GTF      /SAVE THE FLAGS
3644 3771  DCA      FLSAVE      /SAVE THE FLAGS
3645 7501  MGA      /PUT MW IN AC
3646 3770  DCA      MQSAVE      /SAVE THE MQ
3647 6031  KSF      /CHECK THE KEYBOARD FLAG
3650 5261  JMP      C8BY3      /EXIT TO CALL PLUS 1
3651 4767  JMS      CHKCLA      /CHECK LOC 22 BIT 3 CONSOLE BIT
3652 7410  SKP      /ACTIVE CONSOLE PACKAGE

```

```

3653 5261  JMP      C8BY3      /EXIT TO CALL PLUS 1
3654 4766  JMS      XC8TTYI      /GET THE CHAR
3655 4765  JMS      C8GET      /GET THE FLAGS
3656 4764  JMS      XC8CNTR      /CHECK IF CONTROL CHAR.
3657 7000  NOP      /RETURN IF A CONTINUE CHAR.
3660 2241  ISZ      XC8CKP      /BUMP RETURN FOR CALL PLUS 2
3661 4765  C8BY3, JMS      C8GET      /GET REGISTERS
3662 5641  JMP I      XC8CKP      /SAY GOOD BY

```

```

//*****
/C8ECHU
/THIS ROUTINE WILL LOOK FOR A CHAR FROM THE KEYBOARD. STORE IT IN LOCATION CHAR
/CHECK IF IT WAS A CONTROL CHARACTER - SET INMODE - PRINT CHARACTER

```

```

/ C8ECHU = JMS XC8ECH
/EX.  JMS      XC8ECHU      /LOOK FOR CONSOLE CHAR C8PRNT IT
/      /RETURN CALL PLUS ONE AC = CHAR C8TYPED IN

```

/CALLS USED ARE =XC8TTYI=XC8CNTR=C8GET=XC8ECH=XC8TTYI=

```

/ XC8ECH, 0
3663 0000  JMS      XC8TTYI      /WAIT FOR CHAR FROM KEYBOARD
3664 4766  JMS      C8GET      /RESTORE THE REGISTERS
3665 4765  ISZ      INMODE      /SET INMODE IDENTIFYING THIS AS A EXPECTED CHAR
3666 2276  JMS      XC8CNTR      /GO CHECK IF IT IS A CONTROL CHAR
3667 4764  JMP I      XC8ECH      /WAS A CONTROL CHAR - CONTINUE RUNNING
3670 5663  JMS      XC8TYPE      /NOT A CONTROL CHAR C8PRNT IT
3671 4277  DCA      INMODE      /CLEAR FLAG THAT CHAR EXPECTED
3672 3276  TAD      C8CHAR      /GET CHAR IN AC
3673 1275  JMP I      XC8ECH      /EXIT
3674 5663  C8CHAR, 0
3675 0000  INMODE, 0
3676 0000

```

```

//*****
/C8TYPE
/THIS ROUTINE WILL C8PRNT ON THE CONSOLE OR THE LPT WITH DEVICE CODE 66.
/
/ C8TYPE= JMS XC8TYP

```

```

/EX.  JMS      XC8TYPE      /C8PRNT THE CHAR IN THE AC.
/      /RETURN CALL PLUS ONE AC =0000
/      /DO NOT CLEAR THE LINK IN THIS ROUTINE NEEDED BY C8OCT

```

/CALLS USED ARE =C8HANG=XC8CNTR=XC8PNT=XC8CRLF=XC8INQU=

```

3677 0000  XC8TYP, 0
3700 3320  DCA      PNTBUF      /STONE CHAR
3701 1321  TAD      TTYLPT      /CHECK 0=TTY 7777=LPT
3702 7640  SZA CLA
3703 5312  JMP      X00LPT      /DO OUT PUT ON LPT
3704 1320  TAD      PNTBUF

```

```

3705 0046      TLF
3706 0041      TBF
3707 5306      JMP      ,=1
3710 0042      TCF
3711 5316      JMP      C0BY5
3712 1320      X0ULPT, TAD      PNTBUF      /GET CHAR
3713 0666      PSTB      PCLF      /C0PNT IT
3714 4322      JMS      C0MANG      /CHECK KEYBOARD IF HUNG
3715 0662      PCLF      /CLEAR THE FLAG
3716 7600      C0BY5, 7600      /CLEAR THE AC
3717 5677      JMP I      XC0TYP      /EXIT
3720 0000      PNTBUF, 0
3721 0000      TTYLPT, 0

3722 0000      C0MANG, 0
3723 7200      CLA
3724 1316      TAD      C0BY5      /GET CONSTANT 7600
3725 3320      DCA      PNTBUF      /PNTBUF IS NOW A COUNTER
3726 0661      PSKF      /SKIP ON PRINTER DONE
3727 7410      SKP      /NOT DONE YET
3730 5722      JMP I      C0MANG      /SAW FLAG DONE
3731 2345      ISZ      C0CONT      /FIRST COUNTER FAST ONE
3732 5326      JMP      ,=4      /CHECK IF FLAG SET YET
3733 2320      ISZ      PNTBUF      /MADE 4096 COUNTS ON FAST COUNTER
3734 5331      JMP      ,=5      /KEEP IT UP FOR 5 SEC
3735 1764      TAD      XC0CNTR      /GET THE RETURN ADDRESS IN CONTROL
3736 3322      DCA      C0MANG      /SAVE IT IN MANG
3737 3321      DCA      TTYLPT      /ALLOW PRINTING ON TTY
3740 4763      JMS      XC0PNT
3741 3746      MESHANG
3742 4223      JMS      XC0CRLF      /LPT ERROR
3743 4762      JMS      XC0INQU      /PRINT WAITING
3744 5722      JMP I      C0MANG      /CONTINUE TO SAVE ADDRESS
3745 0000      C0CONT, 0      /COUNTER FOR TIMER
3746 1420      MESHANG,TEXT "LPT ERROR"
3747 2440
3750 0522
3751 2217
3752 2200

3762 3435
3763 3103
3764 3200
3765 3424
3766 3072
3767 4200
3770 4123
3771 4124
3772 4122
3773 0212
3774 0215
3775 0260
3776 0007
3777 7774

```

```

4000      PAGE
/*****
/*****

/THIS ROUTINE WILL CHECK LOCATION 22 THE HARDWARE CONFIG WORD.
/TO SEE IF THE CONSOLE BIT 3 (400) IS SET IF SET THEN RETURN
/TO CALL PLUS TWO FOR A ACTIVE CONSOLE PACKAGE AC=0
/IF NOT SET THEN TO CALL PLUS ONE FOR A DEACTIVE CONSOLE PACKAGE.

4000 0000      CHKCLA, 0
4001 7200      CLA
4002 1022      TAD      22      /GET THE CONTENT OF LOCATION 22
4003 0377      AND      (400      /MASK FOR BIT 3 (400)
4004 7650      SNA CLA
4005 2200      ISZ      CHKCLA      /ACTIVE CONSOLE PACKAGE RETURN
/CALL PLUS ONE (1) FOR ACTIVE
/DEACTIVE CONSOLE PACKAGE RETURN
/CALL PLUS TWO (2)

4006 5600      JMP I      CHKCLA

/CBERR
/THIS ROUTINE WILL DETERMINE WHAT TO DO WHEN A CBERR IS ENCOUNTERED
/WILL CHECK IF CLASSIC SYSTEM, WILL CHECK C0SWIT REGISTERS.
/      CBERR= JMS XC0ERR
/EX.      JMS      XC0ERR      /GO TO CBERR CALL IF NOT CONSOLE
/RETURN IS CALL PLUS ONE AC =0000

/CALLS USED ARE -CHKCLA-XC0CRLF-XC0SW-XC0INQU-XC0PNT-XC0OCTA-

4007 0000      XC0ERR, 0
4010 6002      IOF
4011 3322      DCA      ACSAVE      /SAVE AC
4012 6004      GTF
4013 3324      DCA      PLSAVE      /SAVE THE FLAGS
4014 7501      MQA
4015 3323      DCA      MQSAVE      /SAVE THE MQ
4016 7340      CLA CLL CMA      /SUBTRACT A 1 FOR TRUE LOCATION
4017 1207      TAD      XC0ERR      /GET RETURN LOCATION
4020 3321      DCA      PCSAVE      /SAVE ADDR OF CBERR CALL
4021 4200      JMS      CHKCLA      /CHECK LOC, 22 BIT 3 CONSOLE BIT
4022 7410      SKP      /ACTIVE CONSOLE PACKAGE
4023 5263      JMP      NTCLAS      /NOT CLASSIC SYSTEM
4024 4776      JMS      C0GET      /GET THE REGISTERS
4025 4775      JMS      XC0SW      /CHECK SWITCH REG FOR BIT THAT INDICATES
/NO ERROR MESSAGE
/NO ERROR MESSAGE
/IF THIS ERROR MESSAGE IS TO ALWAYS
/BE PRINTED LEAVE AND VALUE AT 0000
/SKIP IF BIT IS 0 PRINT ERROR MESSAGE
/DO NOT PRINT

4026 0374      SETUP1, AND      (0000

4027 7640      SZA CLA
4030 5255      JMP      C0DO10
4031 4773      JMS      XC0CRLF
4032 4772      JMS      XC0PNT
4033 4075      ERRMES
4034 4772      JMS      XC0PNT      /PRINT THE ERROR MESSAGE

```

```

4035 4105      MESPC
4036 1321      TAD      PCSAVE
4037 4771'     JMS      XC8OCTA
4040 4772'     JMS      XC8PNT
4041 4110      MESAC
4042 1322      TAD      ACSAVE
4043 4771'     JMS      XC8OCTA
4044 4772'     JMS      XC8PNT
4045 4113      MESMQ
4046 1323      TAD      MQSAVE
4047 4771'     JMS      XC8OCTA
4050 4772'     JMS      XC8PNT
4051 4116      MESFL
4052 1324      TAD      FLSAVE
4053 4771'     JMS      XC8OCTA
4054 4773'     JMS      XC8CRLF
4055 4776'     C8DU10, JMS      C8GET
4056 4775'     JMS      C8SW
4057 7610      SKP CLA
4060 5273      JMP      C8BY2
4061 4770'     JMS      XC8ING
4062 5273      JMP      C8BY2
4063 4776'     NTCLAS, JMS      C8GET
4064 4775'     JMS      XC8SW
4065 7610      SKP CLA
4066 5607      JMP I      XC8ERR
4067 1367      TAD      (7402
4070 3721      DCA I      PCSAVE
4071 4776'     JMS      C8GET
4072 5721      JMP I      PCSAVE
4073 4776'     C8BY2, JMS      C8GET
4074 5607      JMP I      XC8ERR
4075 0410      ERNMES, TEXT "DMRKD0 FAILED "
4076 2213
4077 0404
4100 4040
4101 0601
4102 1114
4103 0504
4104 4000
4105 4040      MESPC, TEXT " PC:"
4106 2003
4107 7200
4110 4040      MESAC, TEXT " AC:"
4111 0103
4112 7200
4113 4040      MESMQ, TEXT " MQ:"
4114 1521
4115 7200
4116 4040      MESFL, TEXT " FL:"
4117 0614
4120 7200
4121 7777      PCSAVE, 7777
4122 7777      ACSAVE, 7777

```

```

4123 7777      MQSAVE, 7777
4124 7777      FLSAVE, 7777

      $$$

4167 7402
4170 3435
4171 3600
4172 3103
4173 3623
4174 0000
4175 3062
4176 3424
4177 0400

```

0000	11110000	11000000	11111111	11111111	11111111	11111111	11111111	11111111
0100	11111111	11111111	11111111	11111111	11111111	11111111	11000000	00000000
0200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11100111
0400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0500	11111111	11111111	11111111	11111111	11111111	11111111	00000000	00000111
0600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0700	11111111	11111111	11111111	11111111	11111111	11111111	11110000	00000111

1000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1100	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11100111
1400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1500	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11000001
1600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111

2000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
2100	11111111	11111111	11111111	11111111	11111111	11100000	00000000	00000000

2200
2300

2400
2500

2600
2700

3000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3100	11111111	11111111	11111111	11111111	11111111	10000001	11111111	11111111
3200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3300	11111111	11111111	11111111	11111111	11111111	11110000	11111111	11111111
3400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3500	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3700	11111111	11111111	11111111	11111111	11111111	11100000	00111111	11111111

4000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
4100	11111111	11111111	11110000	00000000	00000000	00000000	00000001	11111111

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

A0770	1513	CHKCLA	4000	DSK5B	0156	INMODE	3676
A7007	1512	CHKDAT	0353	USK6A	0146	10T0	0734
ACL	7701	CHKDSK	0337	USK6B	0157	10T1	0741
ACSAVE	4122	CHKER	1112	DSK7A	0147	10T2	0746
ADPOT1	0137	CHKR1	1007	DSK7B	0160	10T3	0714
ADPOT2	0150	CHNPOT	1517	DSKCNT	0105	10T4	0785
ADPT1	1637	CKCOUT	3052	DSKP	0741	10T5	0672
ADPT2	1640	CLDR	0745	DSKSKP	4441	10T6	0727
ADREG	0125	CLKCNT	0132	DTREG	0126	10TCHN	4430
AERRO	1600	CLRALL	4445	ENDBUF	2577	K0003	0060
AGAIN	0533	CMREG	0121	ENDIT	3542	K0007	0062
ALLAGN	0220	CNT	1554	ENDTST	0444	K0010	1324
AMOUNT	0056	CNTRLC	3347	ERHLY0	0736	K0037	0074
APT8	1125	CNTRLD	3400	ERHLY2	0750	K0040	0063
APT8A	4424	CNTRLE	3344	ERHLY3	0716	K0077	1323
AUTO10	0010	CNTRLL	3336	ERHLY4	0707	K0100	1556
AUTO11	0011	CNTRLQ	3300	ERHLY5	0674	K0177	1370
BGN	0200	CNTRLH	3311	ERHLY6	0731	K0200	0066
BGNBUF	0055	CNTRLB	3320	ERMES1	1734	K0212	1226
BGNTST	0127	CNTVAL	3052	ERMES2	1736	K0215	1225
BYRETR	3306	COUNT	1553	ERMES3	2000	K0240	1321
C8BY1	3030	CRLF	4452	ERR1	3536	K0260	0067
C8BY2	4073	CSAVE1	1514	ERRMES	4075	K0277	0065
C8BY3	3661	CSAVE2	1515	ERRQ	0436	K0316	1371
C8BY4	3315	DAREG	0123	ERRON	4437	K0331	1372
C8BY5	3716	DATCNT	0131	ERTX1	1664	K0400	0073
C8CHAR	3675	DATER	1103	ERTX2	1675	K0000	1527
C8CKP	3622	DATOK	1071	ERTX3	1705	K3600	1326
C8CONT	3745	DCLR	0742	ERTX4	1717	K3740	1311
C8D01	3110	DLAG	0743	EXBIT	0120	K4	0061
C8D010	4055	DLCA	0744	EXIT	1506	K4000	0070
C8D011	3407	OLDC	0746	EXITA	3240	K4100	1310
C8D02	3633	DLSC	0740	EXITCK	1551	K5000	0067
C8D03	3150	DMAN	0747	FILCNT	3640	K6500	1571
C8D04	3606	DOCNT	3047	FILLER	3637	K6520	1620
C8D07	3326	DONE	0250	FLSAVE	4124	K7377	1173
C8GET	3424	DONEA	3226	FORMAT	0302	K7400	0764
C8HANG	3722	DOPACK	5012	FRMDSK	0263	K7577	0070
C8RDP3	3466	D0SET	3051	FROCT	1227	K7700	1322
C8RETD	3414	DRIVNO	0100	FRSTUK	1056	K7735	0071
C8RETR	3335	DWST	0745	GOREG2	0117	K7741	1325
C8SETO	3413	DSK0A	0140	GETCH1	3503	K7760	0072
C8SET8	3334	DSK0B	0131	GETUAT	3256	K7771	0557
C8SWST	3545	DSK1A	0141	G0ITA	3243	KAERRO	4426
C8TMP1	3621	DSK1B	0152	G0TOA	3254	KCDF	0075
CAF	0007	DSK2A	0142	GTF	0004	KCNT	1555
CAREG	0124	DSK2B	0153	MEDLST	0553	KERRD	1621
CNTR1	1516	DSK3A	0143	MEDTAD	0552	KILBUF	4435
CHANG	1463	DSK3B	0154	HIGHAD	0103	KLBUF	0752
CHANGR	1477	DSK4A	0144	HITRK	2200	KTICK	1530
CHAR	0101	DSK4B	0155	HOMEMA	0150	KTIME	1557
CHECK	0400	DSK5A	0145	INDEXA	3255	KWAIT	1511

LDA0	0711	PRNTER	4447	TEXT	1652	XRDST	0040
LDA00	4444	PSIE	6665	TICK	4427	XRDTRK	0032
LDA	0700	PSKE	6663	TIME	4425	XREG	0546
LDCM	0720	PSKF	6661	TMPCNT	3546	XRESTR	0033
LDCMO	4442	PSTB	6664	TOCT	1200	XSDKP	0041
LDCUR	4443	PTSTOR	3136	TRKCNT	0104	XTABLA	3257
LDS	4446	QUES1	0232	TSTCHA	3515	XTABLB	3260
LDSB	0134	R0ST	0671	TTYLPT	3721	XTEXT	0545
LDSR	0654	ROSTAT	4440	TWCT	4451	XTICK	0027
LDR1	0611	RECAL	4433	TYPE	4436	XTIME	0025
LDRTRK	4431	RECEIV	4434	UPARUW	3415	XTOCT	0051
LQTRK	2201	REDOSK	4432	UPONE	1215	XWAIT	0034
LWAD	0102	REDOA	3215	WAIT	1527	XWRTK	0031
M10	0077	REDTRK	1000	WASOSK	0242	XXLDSC	0046
M313	0064	RENEX1	0335	WATMBS	3451		
M4	0076	RENEX2	0414	WRKBUF	2200		
MCNTR1	1641	REBTA	1425	WRTOSK	0521		
MES1	2045	RESTER	1443	WRTTRK	0600		
MES2	2066	RESTOR	1400	XAERNO	0026		
MES3	2117	RETRN1	0544	XAPT8	0024		
MES4	2126	SAMAGN	0224	XC8CKP	3641		
MES5	2135	SAVPC	1622	XC8CNT	3200		
MESA	3547	SBCNT1	0106	XC8CNL	3623		
MESAC	4110	S0KP	0740	XC8ECH	3663		
MESFL	4116	SETUP1	4026	XC8ERR	4007		
MESHAN	3746	SETUP2	3025	XC8ING	3435		
MESMQ	4113	SOFT	0136	XC8OCT	3600		
MESPAS	3053	STAEK	1106	XC8PAS	3000		
MESPC	4105	STAOK	1041	XC8PAU	3157		
MOVE	1623	STCNT1	0107	XC8PNT	3103		
MQA	7501	STCNT2	0110	XC8PSW	3456		
MQL	7421	STCNT3	0111	XC8SW	3062		
MOSAVE	4123	STRAUT	0513	XC8TTY	3072		
MULOSK	1153	STREG	0122	XC8TYP	3677		
NEXCHK	0347	SWITCH	0057	XCHANG	0030		
NEXFRM	0276	TABLA	3261	XCLON	0045		
NOSET	3042	TABLB	3271	XCRLP	0052		
NOTOSK	0244	TCHKT	1113	XDOLPT	3712		
NOTEX	0536	TCNTR1	0112	XDOSW	3317		
NTCLAS	4063	TCNTR2	0113	XEND	0135		
NTGO	0474	TCNTR3	0114	XERRU	0057		
OCTEL	4450	TCNTR4	0115	XFROCT	0050		
PASCNT	3050	TCNTR5	0116	XHITRK	0054		
PCLF	6662	TEXAD	1660	XKLBUF	0035		
PCNTR1	0547	TEXCA	1656	XLDAU	0044		
PCNTR2	0550	TEXCM	1650	XLDA	0043		
PCNTR3	0551	TEXDA	1654	XLDCM	0042		
PCOUNT	0161	TEXOT	1662	XLDS	0733		
PCSAVE	4121	TEXEND	2021	XLOTRK	0053		
PNTBUF	3720	TEXEX	1646	XMOVE	0133		
PRINT	1312	TEXGO	1644	XPRINT	0036		
PRN	1252	TEXPC	1642	XPRN	0047		

ERRURS DETECTED: 0

LINKS GENERATED: 132

RUN-TIME: 4 SECONDS

3K CORE USED